



US 20250283139A1

(19) **United States**(12) **Patent Application Publication**
NISHIO et al.(10) **Pub. No.: US 2025/0283139 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **PROMOTER OF ENO1 GENE****Publication Classification**(71) Applicant: **DAIICHI SANKYO COMPANY, LIMITED**, Tokyo (JP)(72) Inventors: **Maui NISHIO**, Tokyo (JP); **Yuto NAKAZAWA**, Tokyo (JP); **Shotaro KAMIZONO**, Tokyo (JP)(73) Assignee: **DAIICHI SANKYO COMPANY, LIMITED**, Tokyo (JP)(21) Appl. No.: **18/860,580**(22) PCT Filed: **Apr. 25, 2023**(86) PCT No.: **PCT/JP2023/016191**

§ 371 (c)(1),

(2) Date: **Oct. 25, 2024**(30) **Foreign Application Priority Data**

Apr. 26, 2022 (JP) 2022-072103

Oct. 6, 2022 (JP) 2022-161649

(51) **Int. Cl.****C12P 21/02** (2006.01)**C07K 16/00** (2006.01)**C12N 9/88** (2006.01)**C12N 15/85** (2006.01)(52) **U.S. Cl.**CPC **C12P 21/02** (2013.01); **C07K 16/00**(2013.01); **C12N 9/88** (2013.01); **C12N 15/85**(2013.01); **C07K 2317/14** (2013.01); **C07K****2317/24** (2013.01); **C12N 2830/50** (2013.01);**C12Y 402/01011** (2013.01)

(57)

ABSTRACT

Provided is a promoter having high foreign gene expression-enhancing activity in host cells such as cultured mammalian cells. An approach for using the promoter to enhance the production of a foreign protein to be used as a protein-based pharmaceutical product is the provision of a polynucleotide that is a Chinese hamster-derived gene promoter, the polynucleotide comprising a nucleotide sequence that is selected from SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13 and SEQ ID NO: 14, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.

Specification includes a Sequence Listing.***pDSLH3.1-hEF1α-Y*****pDSLH3.1-Eno1-Y**

Fig. 1

-pDSLH3.1-hEF1 α -Y



-pDSLH3.1-Eno1-Y

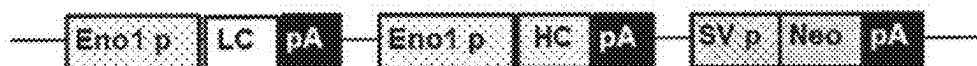


Fig. 2-A

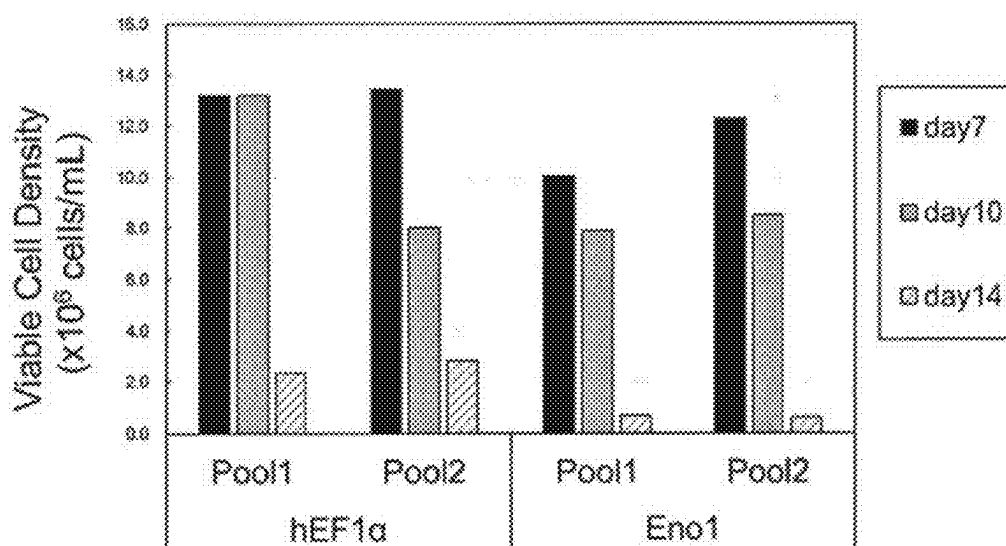


Fig. 2-B

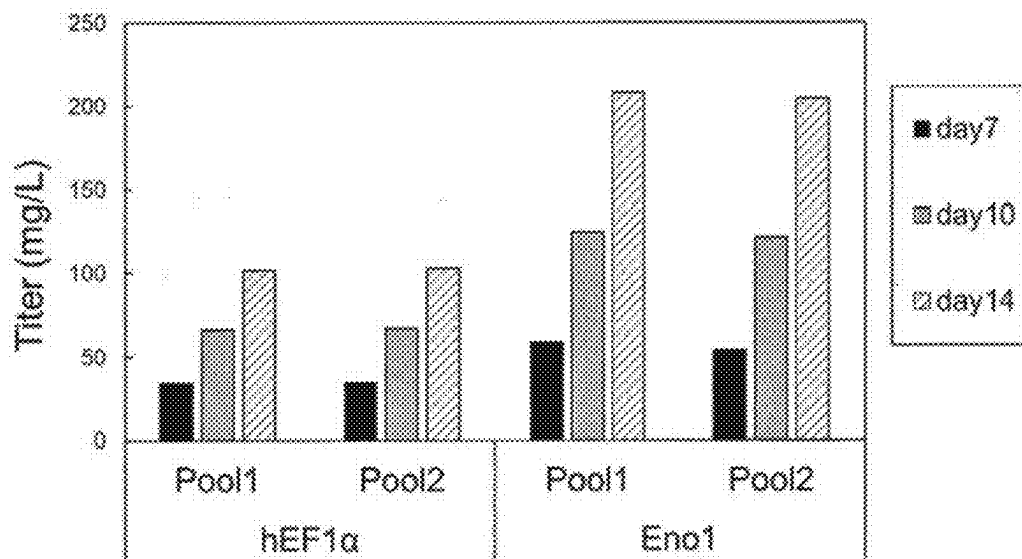


Fig. 3

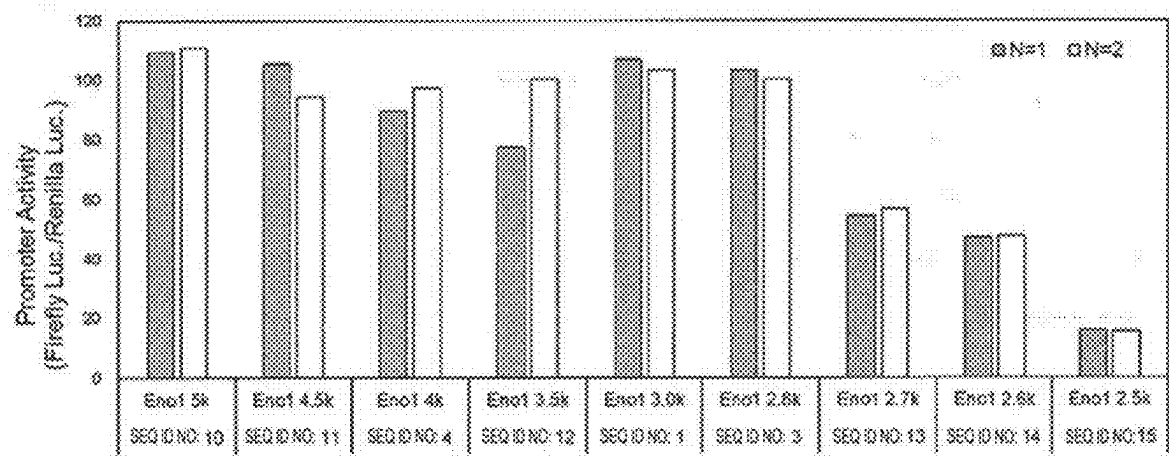


Fig. 4-A

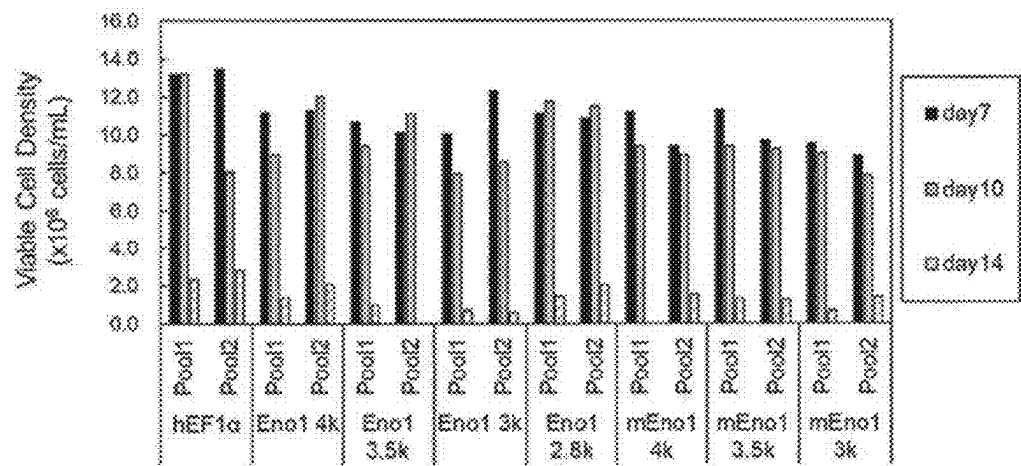


Fig. 4-B

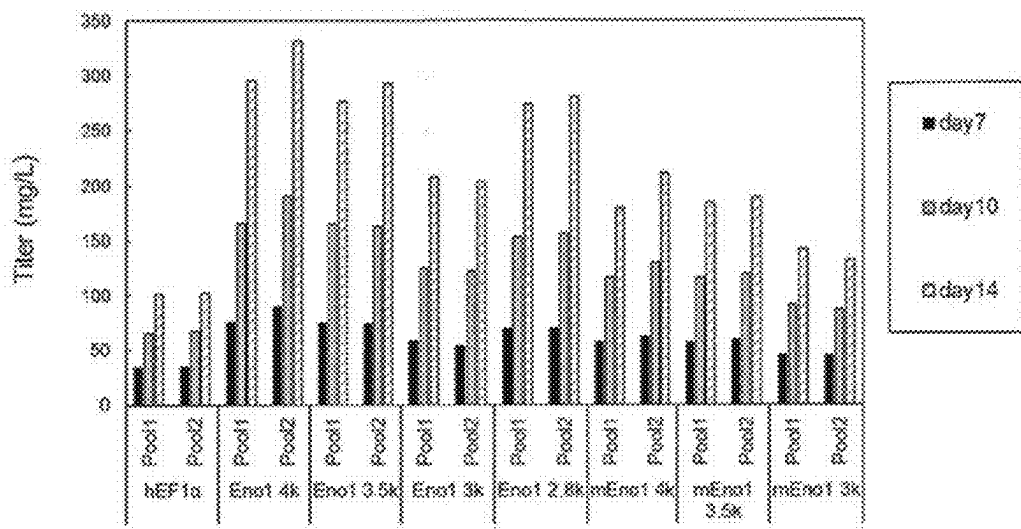


Fig. 5-A

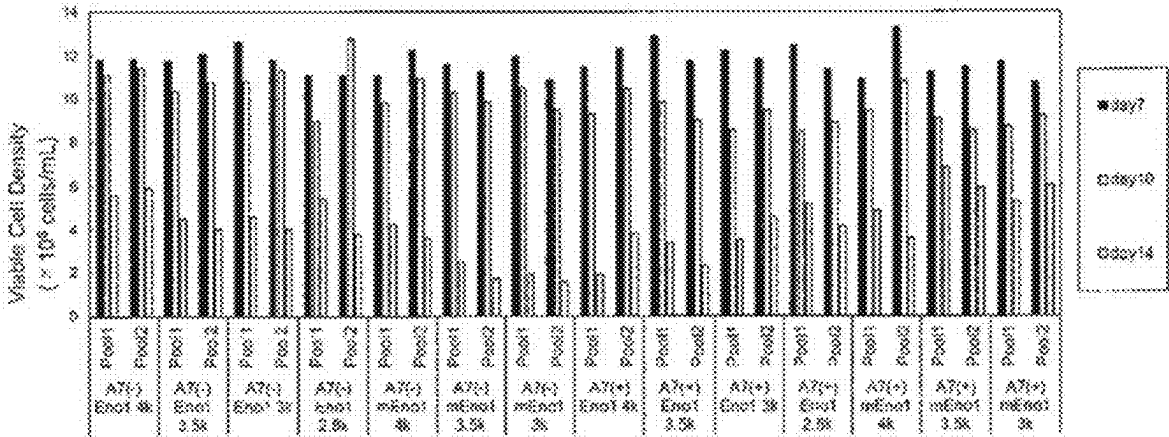


Fig. 5-B

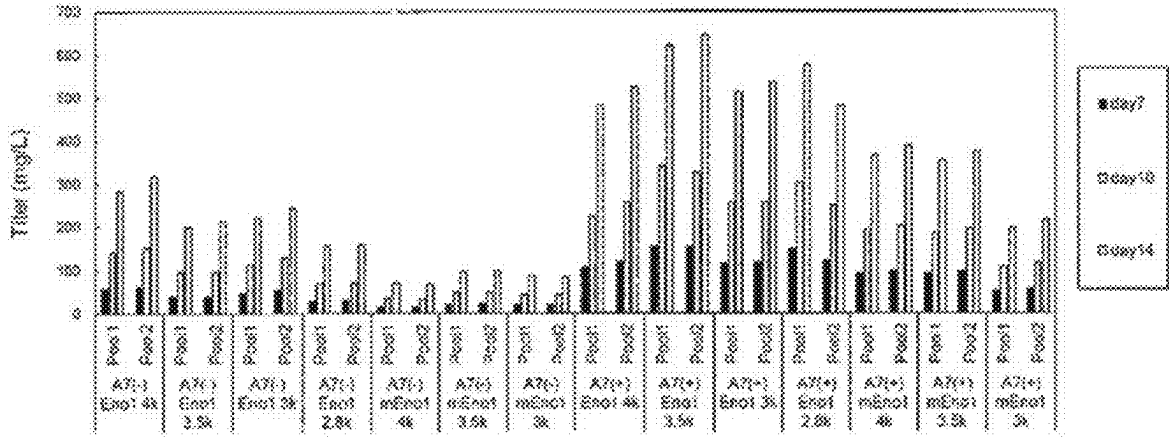


Fig. 6-1

Sequence Listing Information:

DTD Version: V1.3
File Name: PD20-9023WO_230328.xml
Software Name: WIPO Sequence
Software Version: 2.2.0
Production Date: 2023-03-28

General Information:

Current application / IP Office: JP
Current application / Applicant file reference: PD20-9023WO
Earliest priority application / IP Office: JP
Earliest priority application / Application number: 2022-072103
Earliest priority application / Filing date: 2022-04-26
Applicant name: Daiichi Sankyo Company, Limited
Applicant name / Language: en
Invention title: ENO1 GENE PROMOTOR (en)
Sequence Total Quantity: 18

Sequences:

Sequence Number (ID): 1
Length: 3003
Molecule Type: DNA
Features Location/Qualifiers:
- source, 1..3003
 > mol_type, genomic DNA
 > organism, Cricetulus griseus

Residues:

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120	gtgaggctcc	caagctgcgc	cttaacgttt	ccacgcgcct	ggggtctccc	acacctcggg
180	gcggggcgcg	gcgcgaagg	tcaccaccaa	ggtcgagagc	cggacgtgct	cgccgcctcc
240	aagccgagtg	cggtgaggac	tcggagtacg	tgccgcgggc	cggagccgcg	atgtcgggtg
300	cgtagcgacg	cccccgcctc	accagagctc	cgcagacccct	cagctcagcc	cccggctagc
360	aacgggaggg	tcgcccagta	ggccgaaaat	gccacaggcc	cagccaacca	agtctacccc
420	cgcccccggg	ccatcgctct	cagactgtcc	gagccaggag	gacacctcag	tcacgtcttt
480	cttctggagt	gacagcgccg	ggggccaatg	cccggacgga	gaggacagag	ggcgggcccc
540	aaacgcoaag	aatgggcggg	cctcccgag	ctgctcacia	agtcggagtg	ggaggcgctt
600	agaccagctc	gggaaccggc	tgtgtgcgac	gtccctgcta	aacgtctctc	aggggtgtata
660	taggaacctc	tggtcttttc	ctgctcgccc	accggagtgc	caggtaggga	tcgggggccc
720	ctggggaacg	gttcgctcac	gagctccgcg	tggggggccc	cgggtgggtgt	gtgtgcgccc
780	ctccagcctg	ggcgcctgcc	cggggaactc	ggaagcgttc	ggggcagcct	cgtggaggga
840	cgtaggctct	tagagggggc	tggggagggtg	cagatccacc	tgtagcatac	cagacacagg
900	cttgacgggc	aactgggctc	ccaagcctac	aaccgcaagt	gcagaacagg	gtggatcgcc
960	gccccggggg	gttagctaac	tggagggaagg	cttttttttt	tttttttttt	ttttgagggg
	gagaatgatt	tgacctaatg	ctcaggaaaa	accttcacaa	aatggaaggg	cctgaaagaa

Fig. 6-2

1020	ggatgcccaa	gaggcgggtcc	ctcgtcatcg	tccgcctccc	cccccccccc	ccccgggctg
1080	ctcgcagatc	ctgotagggc	tgtgggtcat	ggacogcaat	ctagcgtcac	tgggcgggag
1140	tggiccagcc	catctctgcc	acggaaataa	ctccggtgct	tattctaggg	gaaattgcag
1200	oagaacccct	ggatagotgg	ggagggctta	ggcgacctcg	cccagtttag	gcaagagttg
1260	ogggcttctc	gagtggctgc	tagcaggccg	agaaccgact	taggatgtgt	ttcaggaaat
1320	ttctgtctgc	ccctggogta	agcagcgtgg	tccttttgct	ttogtgggag	gccagagccg
1380	gcctggggcg	atactttgac	ctctaggtcc	actatctctg	ccgggctccg	gccttcagtc
1440	cacttaaaaag	acctagaggg	tgagattttt	atcctgggca	aagacagtgc	agoccaactt
1500	agggatccga	gtcagatgca	tgctggaagg	ggagaaccaa	ctctacacg	ttagcctgac
1560	ttcttagtgc	gcgtggcac	gtggggogtc	ccccggccca	aatagatgaa	aagaaaacta
1620	aagggccatt	gtgccagaca	aggtgatgcg	cacatataaa	ccgggcacag	ggggcgctgc
1680	gacaggagga	tcacttgagt	togggtcaac	ccgtataaca	taggacgtgg	gggtgggaga
1740	agacaagaaa	gtaaatggtg	tttaggcoca	gactgttata	caaaagtgtg	tttgaagtct
1800	acatagtaaa	tcattgggtg	tttaagatgg	aggttttttg	taacatagag	gggtaaaaag
1860	tagatacttt	tottgattag	ggtgtaaaaga	tcctttgtct	cacacaagac	aaagcaacag
1920	aagaagctac	tcagttgaaa	caaaaagatg	cctttgcagc	ttcggccctt	gcgtaccgag
1980	acgtggagaa	ccagttaga	taaagatagt	ttatgaacaa	taatttagtt	tctcacctca
2040	tccaatttat	tgctaagagt	ttttttggta	aaattctgtc	ttggaaggcc	tactggggaa
2100	aaagtaaggg	attttaaaoc	totttagtgt	tgtggccatt	tcttaaaaat	gagctaaaat
2160	gccctttttg	aaagaaatat	gattcagaag	ocatcagagg	atcttaaaata	tggcggtgag
2220	gatggaaacc	attgctgccc	acgtgctagg	caagctagca	gacacgtgcc	agcagaccag
2280	aaagtgtagc	gctgatgato	acagacgaag	gacttaoctg	gtgctgccat	ccaagcccaa
2340	tttctcaaac	taactgaaca	tgcttctggt	gccatgggta	taatataaca	taggotgggt
2400	ggttttttgt	tttgttttgt	tttttgoota	attttacatg	gtttatcctg	tgaccttcac
2460	agcatcatcc	tacttccitt	gcattaggco	ttgagtgtaa	ccagcccatt	gctctaccca
2520	gtcggagaaac	tgatggtttt	agttttgggt	tttttatggg	ttttgcocta	gtggttacaa
2580	aacaggtttt	ttottctctc	tagcaactaca	tgctaagaac	acatttggtta	ccatcccgcc
2640	cgtcttgggg	tttgtatgga	agccagggtg	ttgagcatgo	caottgagca	ctaagccatg
2700	acctagccct	cccacctgct	tttgtacaaa	agcgtagtgt	ttactgattt	gagatgcacc

Fig. 6-3

```

2760 acagtgcact tgggtttggg aacacatttg taggagtggg tttgttgcaac caagtgatca
2820 tcaggcctga ttgggatggc gcctggcctg aattttatgg gatagttcta gaggccctgc
2880 agcagcccac accatcctgc ccactgcttc accgataagt ccagaattcc tcagaagtga
2940 totcttgaaa gtgttaaggt ttgtggaaaa tgtgaattgt tttttttttt tccctacag
3000 aca
3003

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Sequence Number (ID): 2

Length: 3000

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..3000

> mol_type, genomic DNA

> organism, Mus musculus

Residues:

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180 acgggggtcac cttggcagat ggcaccagag agccaggacc agcggcgagg gcctcagtg
240 cgcaggcgca gtatccatgc atttccccct catccccccc acccccaccc aattccacgc
300 gggagtgagg cctcttgccc gcgcatcaac gtttcacac cgcggggggg tocagaacct
360 cccggtgggg ggttggccgg ctggagggcc gagcccgcaa ggatcatcagc aaggctgtgg
420 gccggacgtg tggcgccgca gccacgccga gtgcgtgtga gactcggagt acgtcggcg
480 ggtcaagccc ctcatgcccg ccgocagacc cagcccogcc ccgccccgca tcgggctcc
540 gtagaccctc ggccgggccc caggttocca gccggagggt cggcaggcat tccagctatg
600 ccccaagcag gctaggccca gaaaggcagg acaaccccgc ccccgtgcc ctgcgcccta
660 gtctgggoag gccgagcggc cgcctcggtc acgtcttctt attggagtga tggcacgctg
720 agocaatggg gaggcggcgc ggaggagagg cgggcccga aaggagaaac tgggcgggc
780 ttaacgagca ggaaaggaa acagagtggg aggcgottag tgctgctccg gtacagggtc
840 gcgtcgcgt ctgtccttaa ggctctctc ggtgtccac gcacctctt tcttgcctt
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960 ttgccttggg gaccgggtg gcacgtgcac coactagcct gggcgggagc cgtatccgg
1020 ggagcactcg ggtcagcctc gtggaaggac gcaggctctt ggagggtgga ggtgtgagcc
1080 ccaccagtga tcaccgagtt tcagggttgc aatgtaacc gggtcccaag cagacaacag
1140 caaatgcaga ccacggatga tcggggccgt ggagaataga aaagtacttt ctggtagtt

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Fig. 6-4

1200	aagtgggaagg	tffcagaggg	agggagagca	acttaaaccc	ccgatgctca	ttaaatoctt
1260	tctcagaatg	gaagggccag	aaataaggat	gcccagaggg	cagagcgcca	caccctagta
1320	gatcctgctg	gggtgttagc	tectggaccg	caatcgagcg	tcactggggc	ggagtggloc
1380	agcccatctc	tgtgtggaa	tattctgac	ctgtgagigt	cggtaggta	ctaagagtag
1440	aggaaattgc	agcagtcocg	ttggataatt	gggaagccot	caggagaoc	cgacgcaatt
1500	aggcaagagc	tacaggtttc	ctgggtggct	actagccggc	cgtgaactga	ottaagatgc
1560	atttaagaaa	acttctgtt	cgttctggcc	taggcagtgt	ggttccttgg	ctgactcgct
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1800	aactcctaca	agtgtgaagg	totttgcaga	gcaaagtggt	gcacgtgggt	gtcctccgc
1860	cccataaag	gaaccagcca	tggagatgca	cacatgaaag	tgtagcgcta	gggatgctga
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1980	gaagacaaaa	gagtatggtt	gaggaatagt	aaggttttgt	aacatggtag	gtaaaaagta
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2100	aagtaacttc	tgtggaaaaa	aaagctgcct	ttgcttcagt	gcttgccaca	aaagatgtga
2160	ggaacctagt	tagataataa	tagcctctga	atgctgtatt	gtcttcattg	ctcatocagt
2220	ttagctacag	aatctatttg	gtttctttct	tggaaagcct	gctggggaag	aagtaagaaa
2280	gggacttgca	glttggttgc	aaataatcct	ttcctctcag	gatgttttct	taattaacaa
2340	tctactttga	atacctttta	aaagcgagac	aagggtgcct	tttgaaatgt	actaaccaga
2400	agtcttagct	cttgccataa	tatggcggag	atgaaccca	ggacctccca	cgtgctagcc
2460	agcatggcca	ccaggccaag	aatggcagga	tcattgacca	caggccaagg	agtttggggt
2520	tctggcctcc	tgcctcacac	tttcttactg	gttatcaaat	ctgtgtcagc	tgaatatctt
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2640	ctttgaaatg	tcacatttcc	ctctggtctc	cagggtcagtt	taacctctga	ccttccttca
2700	ctgttgcttc	ccacttctt	ctctgtggag	ccagaacact	gttctatgca	gtcagaggat
2760	ttaatttgcc	cacagtacag	gtagcttctg	agggtttct	gtaggagigg	tttatttcatt
2820	tgtgccatgt	tgacctcaag	cttggcggga	agtccaagcc	accgaacctg	tgtgtaagga
2880	gtagtctctg	ctcccaaac	cctcaotgat	gagtgaactt	cctgtatgtt	acctaacgca

Fig. 6-5

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2940 tagcccatTT ggtacctgtc aagtcacagga cccctcagag gtgatggcct tggaaatgaat
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Sequence Number (ID): 3
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Features Location/Qualifiers:
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      > mol_type, genomic DNA
      > organism, Cricetulus griseus

Residues:
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180  cccagtaggc cgaatatgcc acaggccag ccaaccaagt ctacccccgc ccccgggcca
240  tcgtctcag actgtccag ccaggaggac acctcagica cgtcttctt ctggagtac
300  agcgccgggg gccaatgcc ggacggagag gacagagggc gggcccgaac cgccaagaat
360  gggcaggccc tcccagctg ctacaaaagt cggagtggga ggcgcttaga ccagctcggg
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960  gggctgtggc tcattggacg caatctagcg tcactgggog ggagtgttcc agcccatctc
1020  tgccaaggaa tatactcgg tgcttattct aggggaaatt gcagcagaac ccctggatag
1080  ctggggaggc cttaggcgac ctgcgccag ttaggcaaga gttgcgggct tcttgagtgg
1140  ctgctagcag gccgagaacc gacttaggat gtgttcagg aaatttctg tgcacctgg
1200  cgtaagcagc gtggtccttt tgctttcgtg ggaggccaga gccggccigg goggatactt
1260  tgacctctag gtccactatt ctgcgccggc tccggccttc agtcaccta aaagacctag
1320  agggtagat ttttatcctg ggcaaagaca gtgcagccca acttaggat ccgagtcaga
    tgcctgctgg aaggggagaa ccaactccta cacgttagcc tgacttccata gtgcgcgtg

```

Fig. 6-6

```

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1680      ttagggtgta  aagatccttt  gtctcacaca  agacaaagca  acagaagaag  ctactcagtt
1740      gaaacaaaaa  gatgcctttg  cagcttcggc  ccttgccgta  cgagacgtgg  agaaccagtt
1800      tagataaaga  tagtttatga  acaataattt  agtttctcac  ctcatccaat  ttattgctaa
1860      gagttttttt  ggtaaaattc  tgtcttgtaa  ggcctactgg  ggaaaaagta  agggatttta
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1980      ataigattca  gaagccatca  gaggatctta  aatatggcgg  tgaggatgga  acccatgtct
2040      gccacgtgc  taggcaagct  agcagacaog  tgccagcaga  ccagaaagtg  tagcgtgat
2100      gatcacagac  gaaggactta  cgtggtgctg  ccacccaagc  ccaatttctc  aaactaactg
2160      aacatgcttc  tgttgccatg  ggtataatat  aacataggct  ggggtggttt  ttgttttgtt
2220      ttgttttttg  cctaatttta  catggtttat  octgtgaact  tcacagcatc  atcctaactc
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2460      tggaagocag  ggtgttgagc  atgccacttg  agcactaagc  catgaacctta  gcctccacc
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2700      ctgcccactg  cttcaaccgt  aagtcagaa  ttctcagaa  gtgatctctt  gaaagtgtta
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2807

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Sequence Number (ID): 4
Length: 4000
Molecule Type: DNA
Features Location/Qualifiers:
- source, 1..4000
> mol_type, genomic DNA
> organism, Cricetulus griseus
Residues:

Fig. 6-7

60	atatagtagc	cagttgcctt	ctttcttttt	ttattagtat	tatatttggg	ggttggagat
120	ggaggtcaga	ggacagttga	cagttctccc	tttcagcgct	gggtcctagg	actcgaactc
180	agggtatcag	gcatgatggc	aggtagcctt	gccagagtc	atctcactgg	ctctaagaca
240	gctggitttt	aatggotatc	agtaccttct	tttctttgag	atatggctct	atcattttct
300	tgcctggcct	ggagctcatt	atgtagacaa	ggctagcctt	gctctcaaga	gatctgcctg
360	cttccgtcta	cccagccaga	gccgtgatta	aaccccagat	ttactttctt	gaiggaanaag
420	gttaaatgac	ttcacccggc	ccttccggtc	atttgccctc	tgcctgtgga	atattactag
480	acttotggcc	tacataaaac	cctaaactca	cogtgctttg	gtggctcaca	cctttaatcc
540	cagcacttgg	gaggcagagg	caggcgatct	cagttagtct	gaggccagcc	gggtctacag
600	tacgagttcc	aggacagcct	ctaaaaaat	acaaagaanc	acaaaaacaa	aaccctaaac
660	tctaaacaca	aatacaggcc	tcaaaaatgc	aggtaggttt	gatgtagtct	actaccagag
720	ggaaaagata	tattaaatct	ttcaagaaag	gcaccttaac	ggggaanaag	agtaaatcca
780	cgctgcggtg	gagctggggg	agaaaggagg	gttctaaagg	gtagggtgga	gcagctaagt
840	aaagggtgca	gcctttactt	agcacaagtg	ggaccctggg	ttcgatgcc	tgcacacaa
900	aaccaaata	ttaggtaatg	ttacggggaa	gttcaacttc	ctttggctac	ctctgcgtga
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1260	tcggagtacg	tgcggcgggc	cggagccggc	atgtcggctg	cgtagcgagc	ccccgcctcc
1320	accagacttc	cgcagacct	cagctcagcc	cccggctagg	aacgggaggg	tgcgccagta
1380	ggccgaaaat	gccacaggcc	cagccaacca	agtctacccc	cgccccgggg	ccatcgtcct
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1680	ctgctcgccc	accggagtgc	caggtaggga	tggggggccc	ctgggggaacg	gttgcctcac
1740	gagctccggc	tggggggccc	cgtgaggtgt	gtgtggcccc	ctccagcctg	ggcgcggtgc

Fig. 6-8

1800	cggggactcg	ggaagcgttc	ggggcagcct	cgtggagggg	cgtgggctct	tggagggggc
1860	tggggaggtg	cagatccacc	tgttagcatc	cagacacagg	cttcaggggc	aaotgggctc
1920	ccaagcctac	aaccgcaagt	gcagaacagg	gtggatcgcc	gcccgggggg	gttagctaac
1980	tggaggaagg	cttttttttt	tttttttttt	ttgaggggga	gaatgatttg	acctaattgt
2040	caggaaaaac	cttcacaaaa	tggaaggggc	tgaagaagg	atgcccaaga	ggcggtccct
2100	cgtcacgctc	cgcctccccc	ccccccccc	cgggtgctc	gcagatcctg	ctagggtgtg
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2220	gaatatactc	cgggtgcttat	tctaggggaa	attgcagcag	aacccctgga	tagctgggga
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2460	taggtocact	attctgcccc	ggctccggcc	ttcagtcac	ttaaaagacc	tagagggtga
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Fig. 6-9

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3660 cactacatgc taagaacaca ttigttaacca tcccgcocgc tctgggattt gtatggaagc
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3780 gtacaaaagc gtagttttta ctgatttgag atgcaccaca gtgcacttgg gtttggaac
3840 acattttagt gagtgggttt gttgcaccaa gtgatcatca ggcttgattt ggatggcgcc
3900 tggcttgaat ttatgggat agttctagag gccctgcagc agcccacacc atcctgccc
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Sequence Number (ID): 5

Length: 8450

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..8450

> mol_type, genomic DNA

> organism, Homo sapiens

Residues:

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180 aattcaataa gacaaccocaa tttaaaaaca ggcaaaggat gigaataggo atttctocaa
240 agatacgga aacggccaa taagcacata aaaagatgct caaaatcatt tgccatttgg
300 gaaatgcaat caaaaccaca atgaggatct acttcacgcc cattagggtg gctatagatc
360 agaaagtcag ataacatgtg ttggcaagca catggaaaca ctgaagtctt tacacactgc
420 tggtaggaat gtaaaatggt gcagcoactg tggaaaacag ttttcaatt tctcaaatg
480 ttaaacacag ttatcataca cccaagcaat tctactotta ggtatatacc caagagaaat
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720 gagtgaatct tgaaaacact atgctaaatt aaagaagcca gtcacaaaag gccgtgtatt
780 gcatgatttt atatatacat gaacttttat atatataata ttatatatat tatatataat
840 tttatatata taaatttcta tatataaata tataaatca tatatatgat atatattttt
900 tcatatacat catatatatt tacaaaaatt atatatacata tatcatatga tatatgagat
atatatcatg atatatatga tatatgatat atatcatatg agatatatga tatcatgaga

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Fig. 6-10

960	tataatgat	catatgat	ataatgat	gataatcat	gataatata	taataat
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1380	gataatcat	tataatgat	agatat	gataatcat	tataatgat	agatat
1440	gataatcat	tataatgat	atcatata	tacacata	atcatata	catcatat
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1560	gaagtata	agtgttgc	acagcgtg	aggagcagg	aagtgtgag	gactgctaat
1620	ggatatggca	cttttttgg	gggtgtga	aaatgtctg	gtcagaca	ggcaattaca
1680	aaactgtata	cacacgaaa	accaagaat	cacacactt	aaaaggagg	atttagctcg
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1920	ccagttcac	atgcataaac	actgttaaac	ataaattatc	tccaagtaat	ctagaagaa
1980	aatgagcaca	taagacgtct	tctaataaca	cacatatatt	totttacatg	ttacatttaa
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2520	tctaaccatc	cacaacccta	agctggtctc	tcatcaaacg	ggtacttttc	accacccaaa
2580	ttcaattaat	tcactcttac	aataatgaag	aatagtcgoc	tacagctac	cttttcagc
2640	cttgattcaa	tcatttatca	attttatctt	caaagtccct	cacttcaggg	agatgatata

Fig. 6-11

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3240	catctaaaga	cctagggtgc	aagcatagct	ctatgagtac	aatcccgtgc	ctgagattac
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Fig. 6-12

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4800	atgaaagaac	ctatacattt	atggaccaat	ctacottaac	taacttgtca	agtgttccctg
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Fig. 6-13

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Fig. 6-14

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Sequence Number (ID): 6

Length: 8420

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..8420

> mol_type, genomic DNA

> organism, Homo sapiens

Residues:

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720      agggcctgga ttttacaaaa gtaacacaat gaagaagaga gatgcaaggo tatttgaggx
780      tgtttggtgg agagattgta aaatattago taagtaagaa ggggactgca aattttagtg
840      gtataaagga atgaggaaaa gtgtaaatac agtggggtca aagaatgttt ggagccaagg
900      caotagaggo aattagctga aaatgtaggt gattattggt gagtgcacatg gtttaaatga

```

Fig. 6-15

960	aaagtataga	agggtacaat	tatccatcat	gaaaagtict	agggtacaac	taagatctga
1020	gtagctgaag	tagaatgaaa	gtagnaigga	ccittccata	tccagccagg	ttcagtgaca
1080	gaaggtagg	aaacaaatta	taaaccactt	gagagaacat	atcccctaag	ttgtttttgc
1140	tatttttctt	tcagcatata	tttgittgaa	tgcacaactat	gttcagttca	attaatatgg
1200	gctttctaaa	taagggtctc	agcactggat	aatcctgcc	tttattttga	tacattcoat
1260	cctgtctctc	agatctattg	gcactctacag	gatgtctttt	gagaagatgg	gcattcacat
1320	ccctaigtcc	tagcaaat	ccaactcaga	aaaccacatt	aggcttctct	atatactctc
1380	caactatttc	aattggaaaat	acaattctct	gatttcttcc	tatgatattt	atcaaaagaga
1440	atggtgctg	ccagttctag	ggtgggggaa	ctcaatacaa	atcaccaacc	tttagatgac
1500	accctgtctt	caaagtgtt	tcaaagtctg	gcagaaaaaa	agtaaccagt	ggctataaga
1560	ccaccagga	gttcagtc	gcattctaa	tagcagatca	ctggaaatga	attggctagt
1620	gagttcattt	tactcttctc	ttcttggcca	catgttacgg	cccttgtaac	ctgcacgttc
1680	tctttccag	acttacaag	catgttctct	tgaattcgtt	ctctttttta	attcacacag
1740	tcttaatgat	tcttctttca	caagagtctt	tcactcttca	aattcagttc	aagtcaccca
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1920	ccctaattcc	ttggctaggc	cagaacagaa	tattgtctaa	aaactcctca	cgtaagcagt
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2040	gtgcatactt	atagggtaca	atctgatgtt	tcatatgtg	tttaaatggg	tgcattgtgt
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2160	tgcgaagaag	caggaatttt	tagcagacaa	ctgagatgct	tottgttcac	actaagtc
2220	tctgacgatg	gatttacata	acttgttgtt	tttttttgtt	gtgtgttttt	gagacagagt
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2340	ctcccgggtt	caaacgatto	tcttgcctca	gctcctgag	tagctgggat	tacaggtgca
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2460	cctgtgtgtg	tcaaatctgt	ggcctcaagt	gatctaccag	ctgcggcctc	ccaaagtgc
2520	gggattacag	gtgtgagaca	ccaagcctgg	tacatttaca	tttcttatct	ggatctttcc
2580	tttagtaagt	gctaaggaa	cttacttccc	ccaatatttt	ttcttatctc	aagtgtttag
2640	catgtatcat	gttactactt	tgcagacatt	tgattttccc	ctttgtttac	tgtaaagtat

Fig. 6-16

2700	atTTTTatag	cotTTtgtaat	agaagtatto	taaaaToTgc	ctgcaacctA	totTTctgac
2760	tctgcatttt	agggAataat	tctctgttgt	ggaatgaaaa	aaaaaacaga	gootgtggag
2820	tcagagatct	catttcaaat	tatagttatc	cctaggaata	aatctgagtg	acaggtagta
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3180	agaagocitt	ccagatcatc	ccatggccaa	gaacgatcgc	tgttctctct	tctttacata
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4200	tcatttctgt	ccaatgggag	aaacattttc	gtttgagacc	tccgtgaata	atacagctct
4260	ttagttagga	gagctgcatt	ttgagtggtg	caggcagaat	ggcgatctct	caccacaca
4320	aacactaaga	tagagagaga	cagagacaga	gacagagaca	gcagagagag	acagagaaag
4380	gaagtacagg	tactcagata	gagataagcc	atttcttgac	attaagaaat	aaagtagaat

Fig. 6-17

4440	ccattggagg	gaataaaaac	tgocctcagga	acagagittaa	ttcacataca	catgcaggta
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4560	gtgtgtacat	tcagocctcc	atatccatgg	attttgcatt	cacagattca	accaaccatg
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5100	catatatgtg	tgtatgtgta	tgtgtgtgtg	tgtgtgtgtg	tgtgtgtatg	ctgtctttcc
5160	tcggtatcac	agggaaattg	agatatatat	attcttttca	gtacaaaaaa	aattgaacac
5220	agatgggtat	ggtaccagaa	cagaaggtaa	agacacatga	aaaaaatttg	caacaacatg
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5820	ggtacaaata	ttttttaaaa	aacaaataat	tatttaattc	gccttttaaa	agctttttaa
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5940	cagagotcca	tatatgtggo	tatgatagcc	ttaacctgatt	cttgccaaga	atctagtgcc
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6060	acctgttgaa	atatgacctg	gcgcaaacac	ctatgccact	tgcttatgaa	atcatatagg
6120	ttttcggtgt	gcagttttga	ctgaatgagg	gagtttaago	tggaccacaa	ggggcccoct

Fig. 6-18

6180	ctgtcaataa	cgtactccat	ttgtgtatta	agtcaaaaaat	gaaatggaag	agaaaagaaa
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6300	aagtactctg	cccttttaag	aatcttgcac	tcacattcct	gatgaagtta	ttttctctcc
6360	tctcaactgat	tcccatttca	ctctattaca	tagcacccgtg	ttccccagga	gctctgaat
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6480	taggcctatgg	gtattccttt	ttactgaagg	tatgacatat	agctgccag	gocctgactaa
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6660	atcccacata	caaccacagt	aacttcaatt	tacatgtttt	attctgttac	caaactaaat
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6840	attatgtgtt	ttcagtgcac	taaaccaactg	gtctgatacc	tagcccaaca	ttctattaaa
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7680	aacttctcaa	actatgtttt	gaccggctag	cataatgttt	cagtctttct	ggagaatgoc
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7800	gggaagaaca	ggaagaggac	agatcaaatt	actcaagagg	aaggacaaga	aataaggnaac
7860	caaattatca	acaattggag	aaagaaagct	gatgtcagta	tcatttcata	tatgattatg

Fig. 6-19

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8220   tttccaaaga  gtccattaaa  agatatagaa  gaagggaggc  agcaataaca  ctaagagaaa
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8340   ctttaacaaa  aattttaagt  tgcaatagat  gttcaacttt  aaatccatcc  cagaaaaatt
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Sequence Number (ID): 7

Length: 8475

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..8475

> mol_type, genomic DNA

> organism, Homo sapiens

Residues:

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Fig. 6-20

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1920	cctttagtga	tcttcacttt	tatggctgct	tccgcctaag	aagttcatca	tctgtactta
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Fig. 6-21

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4260	atcaacagtg	ttttaacagt	ttttaagcaa	aagtcagctt	tgtttatcta	agattttttt
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Fig. 6-22

4380						
4440	ttcgacaggt	accacccctot	aacataactg	aaataatgtc	taccattact	ggatcttgc
4500	agcaagaat	ctcaaatitt	cccacttgg	tgtaaattat	tttgtaatct	ctagtgttta
4560	aggtgogctt	gtcctatcta	atccoctccc	tggcaggaca	ccttacagaa	octacccctt
4620	acactagtca	ttaagcacca	tcagggaagg	atggctgtgt	cactggctctg	tttggttatc
4680	octactgac	ctaccatgtg	gtgattatct	atgaottccc	taatccctgg	ctgccttagc
4740	tgggaactgg	tgacatgctt	ctcaggttgc	ogctggcttt	acagtccctt	actgcccattg
4800	ccactttgga	gataggcagg	gctagtaott	ttctatataa	gcccccaaac	ttgactttgt
4860	gtttcacagt	aggtgaaaaa	gttgggtctc	ttttctttta	cttttctttc	cacaagatga
4920	taaagotagg	ggaagcctgt	ggaocatggt	tatttctgca	actgcaatga	ttgatitggg
4980	cttctgtctg	cttacttctt	aaactttgtg	ctcagtgtca	gatccctagc	agtttctatc
5040	ccctgctctg	ctaaaaaaga	atggatgttg	actctcaggg	octagtctct	tttaattaaa
5100	ttgtattttt	gttatcattt	ttattattat	tattttgaga	tggggctctt	ctctgtcgcc
5160	caggotgaag	tgcagtggig	caatcacagc	tcactgtttt	agootcctga	gtagctggga
5220	ctacaagcgt	catgocacca	tgcttctttt	taatttttta	aaatggtttt	ctgccttcaa
5280	ttctaagcac	ttctcaattg	taaccaagag	ataatacttt	ttatgaattc	ttaaagttat
5340	caacagatac	tcaaagtttt	agcaaagtct	aaatgatatt	aagcttgctc	ttattgcccc
5400	agtgaacttca	atgactattt	gttaattgca	accaagggtc	attttttaaa	tgaatataa
5460	ttattattat	atatataata	ttagggtcct	caaataccta	aaagtttagc	aaaaatcaaa
5520	taatattgtg	catattcttt	tattactgta	ttagtcogtt	ttcatgttgc	tgataaagac
5580	ataccaaga	ctgggcaatt	tacaaaagaa	agaggttcac	tggactcaca	gttccaogtg
5640	gtgaggagg	octcacaatc	acggcagott	acgggattgt	tgagaaatga	cacttctcaa
5700	gtgagggcta	aaatatctct	gtggtagttg	ttctgattca	agtattgaat	tggttttttt
5760	tgtttttttt	gagatggagt	ttcgttcttg	ttgcccaggg	tggagtgcac	tggcagcatc
5820	tcagctcacc	gcaacctctg	cctccgggtt	tcaagtgatt	ctcctgcttc	agootcccaa
5880	gtagctggga	ctacaggcat	gagccaccac	aoccagctaa	ttttgtattt	ttagtagaga
5940	catggtttct	ccatgttgg	caggctggtc	tcaaaactcc	aacctcaggt	gatccacctg
6000	ccttggcctc	ctaaagtgt	gggattacag	gcataagcca	ccgtgcccg	ctggagcatt
6060	ggtatataaa	agctgcctag	gtaaactctaa	cotttggccc	catacatctg	aaggatacct
	acaatgcacc	tgaanaatgc	aactgaaaca	gtagttccct	gggaccacac	actcagaag

Fig. 6-23

6120	ggggtgtatc	aggagatcta	gggaccagga	gggtggaaga	cctaaggcag	cactacagat
6180	gatggagaaa	aaoccaactg	ggaggggcga	tcctaaccott	gagaatcaact	gagatcatgc
6240	agaagtattt	gatcctacag	cattaatatt	gtattgtatt	gtatttagtat	atatatatag
6300	tgtatatata	tagtattagt	atatatatig	tattgtatta	gcataatatat	actaattgia
6360	tigtattgta	tttatatata	tagtattgta	ttagtatata	tatacagtat	atatgtatat
6420	atactaatac	aatgtactaa	tacaatacaa	taccatatat	atatacacta	acacaataca
6480	attagtatat	atatatatat	atatactaact	acaatacaat	actatatata	tactaataca
6540	atatatacat	atatactcac	caagacatat	tagtggcttg	atgtctggct	gccacactca
6600	tcttctacct	tcagctctgc	totaccaaat	atcatttgtt	tctgggatct	ttgcagtcca
6660	aggaaotcca	tccttgatat	cccacocctt	actaactttt	tttttttttt	tttttttga
6720	gacggagtct	cgtgtgtca	cccaggctgg	agtgcagtg	tgtgatctcg	gctcactgca
6780	agctccacct	octgggatca	caccattctc	ctgcctcago	ctcccaagta	gctgggacta
6840	caggtgccc	ccaccacacc	aggctaattg	tttaccgtgt	tagcaaggat	ggtctcgatc
6900	tcctgacctc	atgatccatc	cgccttggcc	tcctaaagtg	ctgggattac	aggcataagc
6960	caocgcaccc	ggccacccct	tactaatttt	tagtaacgtc	caaggattaa	aggaaatttg
7020	ccttacctat	ttaacaggaa	tcaacagggt	taactcact	ccctttctaa	aaataattta
7080	taaacattgc	agacaatctc	atctatocct	gtctaaaactg	tgtggaatta	ctgccattta
7140	atgtaatoag	totactcatt	tagtttgcct	aaggaatttt	tgaaaaaaca	gttaaatgaa
7200	tgaottaatg	gaataaccag	gaagttgaag	tcctcaantag	taagaatgaa	ctcttgcctt
7260	ctggataatc	aaatgggtcc	ttcctccttc	aggtagatca	tgccatttcc	tcacttacac
7320	tgaacaggta	aacaacataa	ttactgaott	caacttctag	ttaattcctt	cttttatcac
7380	tgagtatcct	ttggctggga	gttttgttgg	ctatgctgcc	attttttcta	gttatcacag
7440	tcctataaca	taocaaatcct	tcaatataac	tcactcttaa	atttgtgttt	taoottotca
7500	agaagttatt	aattatgcc	gtgctaaatc	ttctaaaatg	attgttgact	tgttgattag
7560	cccccattga	attccccctt	cccgcccttc	agcacgtaag	gaatggccct	ttgcttactt
7620	ccacagatcc	ttaaatctac	cagttagaag	ctaatagcct	acctctctac	caggaaggaa
7680	ctgtgggctg	gaacataata	catgttgact	tataatttct	tagaaaattg	tgtgagaaac
7740	atcaaaactcc	tgattccagg	atatgccaaa	gacacatcat	taaaaagcaa	aacaaaacaa
7800	aacaaaactc	atttgacgtt	gctagttagt	gcataattca	tcangatcag	ctcaaatcaa

Fig. 6-24

```

7860   tagaagtgag attttcacac aaattagact gtagtgcttt tttttttaac ttatctttac
7920   catatgattt ttaacggtaa aaaaaatcgt ttgagatatt agatgtataa tatttatcat
7980   ocaattactt cattagttca atcttttttc aatggcgctc ctgcacttga gaataaggto
8040   agaaaatttc atgttctgat ttcatgctga ttttcagaag aaaaatgtta gttttgtata
8100   gaataaccca tctaagaaa tacatttctt attatatttc ttatcttata tttcttagga
8160   caatgagota ttcaaagggt gatgataacc agcaccatca gtcagcatta tctaagaata
8220   agaatctgtg tttctacata cagaacctct aaaaaggaac ctacacttaa caggattccc
8280   caggcaattt ggatgcacat taaagcttga gcaacactgc attagaaagt tagttttoca
8340   tcacaaaaac agtaacaaaa ggaatataaa gtaagttact ttaantaatat aagaagaggg
8400   gcaggccggg cgcagtggtt caogcctgta atcccagcac ttggggaggc tgaggcgggt
8460   ggatcacctg aggtc
8475

```

Sequence Number (ID): 8
Length: 28
Molecule Type: DNA
Features Location/Qualifiers:
- misc_feature, 1..28
> note, Primer K24
- source, 1..28
> mol_type, other DNA
> organism, synthetic construct

Residues:
ttcgccggcg cggcagcgag cgccagg

28

Sequence Number (ID): 9
Length: 34
Molecule Type: DNA
Features Location/Qualifiers:
- misc_feature, 1..34
> note, Primer K25
- source, 1..34
> mol_type, other DNA
> organism, synthetic construct

Residues:
ttcactagtt gtctgtaggg gaaaaaaaaa aaac

34

Sequence Number (ID): 10
Length: 5000
Molecule Type: DNA
Features Location/Qualifiers:
- source, 1..5000
> mol_type, genomic DNA
> organism, Cricetulus griseus

Residues:

Fig. 6-25

60	gttcttttaga	cgttggttact	catcttctct	cocggtgcct	ttgatctgcc	actgccatac
120	aatgcgccaa	ccagcgccctg	caagtgagaa	aagacgtata	aatgcctgtg	acaaggcaaa
180	cagccctgtg	ctggcttcag	acaaatgtga	gcagggcctg	taacagccctg	gagagcccttg
240	gcaacagggc	atgtggctgc	ttgtgaatto	tgagacatcc	tggaatcttg	caaaggcagc
300	tgcttgatac	catgttgctt	tagattccgg	cctgagacac	ciggtctatt	ggaataagag
360	ttaicttact	ctatctttat	ctctccaacc	attttagtgt	aaaacaaaga	ggctgtctgg
420	agccctctca	ggcccatagt	cttggaatca	catggaaaaac	agagctcaga	gtgtccact
480	tcttacaagg	tttgcaaaag	ggagagtgtg	tgcttcagct	gotgagtcgt	gagactgcag
540	aatattccgt	tcatgcagcc	aacacacatc	agcatattat	tggtgaaatc	agactgcttg
600	aaatccctaa	catctgttaa	tgaatcatct	tacaatttcc	ttgttgcttc	tttgttgaat
660	ttaatcccaa	agctcttgct	tcttcactca	tggaaactta	acaccttgga	ctcaataagc
720	cttcagtgtt	acccaaggcc	tggaaacct	tggtcaggaa	ctgcctggag	caaatgcta
780	gggtttaaga	gagaaatgca	acttccigta	gctgactaga	tgatagaata	accaaagcaa
840	aaaaacatat	ttcaatctct	ctatgaaaga	cgatatggaa	aatatagaaa	caagagacat
900	tggcttccag	acttacagcc	cagaagaaac	agaaaataca	accggaagag	agggtctgac
960	ttcagccctc	ccatgaagaa	atatgtgcac	tccagagact	tgcagattct	tgtagtgttt
1020	gagacaggat	cgtctccctt	ggtagcccaa	gcttgccctc	atatagtagc	cagttgcttt
1080	ctttcttttt	ttattagtat	tatatctggg	ggttgagat	ggaggtcaga	ggacagttga
1140	cagttctccc	ttcagcgct	gggtccatagg	actogaactc	agggtatcag	gcattgatggc
1200	agggtccctt	gccagagtc	atctcaactg	ctctaagaca	gctggttttt	aatggctatc
1260	agtaaccttc	tttctttgag	atatggtctc	atcattttct	tgcctggcct	ggagctcatt
1320	atgtagacaa	ggtagacctt	gctctcaaga	gatcigcctg	cttcogtota	cccagccaga
1380	gccgtgatta	aacccagat	ttactttctt	gatggaaaag	gttaaatgac	ttcaccgcc
1440	ccttccggtc	atttgcctca	tgcctgtgga	atattactag	acttctggcc	tacataaaac
1500	cctaaactca	ccgtgccttg	gtggctcaca	cctttaatcc	cagcacttgg	gaggcagagg
1560	caggcgatct	cagttagctc	gaggccagcc	gggtctacag	tacgagttcc	aggacagcct
1620	ctaaaacaat	acaaagaaac	acaaaaacaa	aaccctaacc	tctaaacaca	aatacaggcc
1680	tcaaaaatgc	aggtaggttt	gatgtagtct	actaccagag	ggaaaagata	tattaaatct
1740	ttcaagaaag	gcaccttaac	ggggaaaagt	agtaaattca	cgtgcgggtg	gagctggggg

Fig. 6-26

1800	agaaaggagg	gttctaagg	gtagggtgga	gcagctaagt	aaagggtgca	gccttiactt
1860	agcacaagtg	ggacootggg	ttcgatgcc	tgcacacaac	aaacaaatga	ttaggtaatg
1920	ttacggggaa	gttcacttto	ctttggctac	ctctgcgtga	ggctctccgc	agcaaggaca
1980	aaaggacagg	gtcaccttgg	cgggtggcac	ccgacagcca	ggaccaggat	ccgogggggt
2040	gggggggggt	gggggtgggg	ggcagcgagc	ggcagggcgc	gggtgtccac	accccccctc
2100	ccgcgcgcgc	ttccacgcga	gtgaggctcc	cacgctgcgg	cttaacgttt	ccaogccgct
2160	ggggtctccc	acacctcggg	gcggggccgg	gccgcgaagg	tcaccaccac	ggctgagagc
2220	cggacgtgct	cgccgcctcc	aagccgagtg	cgtgtgggac	tcggagtacg	tgcggcgggc
2280	cggagcccg	atgtggctg	cgtagcgagc	ccccgcctc	acccagactc	cgcagaccct
2340	cagctcagcc	cccggttagc	aacgggaggc	tcgccagta	ggccgaaaat	gccacaggcc
2400	cagccaacca	agtctacccc	cggcccccgg	ccatcgtcct	cagactgtcc	gagccaggag
2460	gacacctcag	tcacgctctt	cttctggagt	gacagcgccg	ggggccaatg	cccggaaggc
2520	gaggacagag	ggcggggccg	aaacgccaa	aatgggcggg	ccctcccag	ctgctcaca
2580	agtcggagtg	ggaggcgctt	agaccagctc	gggaaccggc	tcgtgcgcac	gtccctgcta
2640	aacgctctcc	aggggtgtga	taggaacctc	tgtcttttt	ctgctcgccc	accggagtgc
2700	caggtaggga	tcggggggcc	ctggggaaag	gttcgctcac	gagctccgcg	tggggggccc
2760	cggtaggtgt	gtgtgcgcgc	ctccagcctg	ggcgcgtgcc	cggggactcg	ggaagcgctc
2820	ggggcagcct	cgtggaggga	cgtgggctct	tggagggggc	tggggaggtg	cagatccacc
2880	tgttagcatc	cagacacagg	cttgcagggc	aactgggctc	ccaagcctac	aaccgcaagt
2940	gcagaacagg	gtggatcgcc	gcccgggggg	gttagctaac	tggagggaag	cttttttttt
3000	tttttttttt	ttgagggggg	gaatgatttg	acctaatgct	caggaaaaac	cttcacaaaa
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3120	ccccccccc	cgggcctgct	gcagatcctg	ctagggctgt	ggctcatgga	cgcgaatcta
3180	gcgtcactgg	gcgggagtg	tcagcccat	ctctgccacg	gaatatactc	cgggtcttat
3240	totaggggaa	attgcagcag	aaccocctga	tagctgggga	gggcttaggc	gacotcgccc
3300	cagttaggca	agagttgcgg	gcttcctgag	tggctgctag	caggcogaga	acgacttag
3360	gatgtgtttc	aggaaatttc	ctgtcgcccc	tggcgtaagc	agcgtggctc	ttttgttttc
3420	gtgggaggcc	agagccggcc	tggcgggata	ctttgaacct	taggtccact	attctgcccc
3480	ggctccggcc	ttcagtcac	ttaaaagacc	tagagggtga	gatttttato	ctgggcaag

Fig. 6-27

```

3540 acagtgcagc ccaacttagg gatccgagto agatgcatgc tgaaggggga gaaccaactc
3600 ctacacgtta gccitgaottc otagtgcgcg ctggcacgtg ggcgctcccc cggcccaaat
3660 agaigaaaag aaaactaaag ggccattgtg ccagacaagg tgatgcgcac atataaacco
3720 ggcacagggg gcgtgcgcac aggaggatca cttgagttcg ggtcaaccog gtatacatag
3780 gaogtggggg tgggagaaga caaagaagta aatggtattt aggccagac tgttatacaa
3840 aagtgtattt gaagtotaca tagtaaatca ttggtgtttt aagatggagg tttttgtaa
3900 catagagggg taaaagtag atacttttct tgattagggt gtaaagatoc tttgtotcac
3960 acaagacaaa gcaacagaag aagctacica gtigaaacaa aaagatgcot ttgcagcttc
4020 ggccottgoc gtaogagacg tggagaaccc agttagataa agatagtita tgaacaataa
4080 tttagtttct caoctcatcc aatttatlgo taagagtttt ttgtgtaaaa ttctgtottg
4140 gaaggcctac tggggaaaaa gtaagggatt ttaaacctct ttagtgttgt ggocatttct
4200 taaaatgag ctaaaatgcc ctttttgaaa gaaatatgat tcagaagcca tcagaggatc
4260 ttaatatgg cgtgaggat ggaaccatt gctgocacg tgotaggcaa gctagcagac
4320 acgtgocagc agaccagaaa gtgtagcgt gatgatcaca gacgaaggac ttacgtggtg
4380 ctgccatcca agcccaattt ctcaactaa ctgaacatgo ttctgttgcc atgggtataa
4440 tataacatag gctgggtggt tttttgtttt gttttgtttt ttgocsaatt ttacatggtt
4500 tatcctgtga ctttcacagc atcatcctac ttcttttga ttaggcottg agtgtaaoca
4560 gccattgct ctaoccagtc ggagaactga tggttttagt ttggggtttt ttatgggttt
4620 tgcocotagt gttacaaaac aggttttttc ttctcottag cactacatgc taagaacaca
4680 tttgttacca tcccgccgcg totgggattt gtatggaagc cagggtgttg agcatgocac
4740 ttgagcacta agccatgacc ttagcctccc aactgotttt gtacaaaagc gtagttttta
4800 ctgatttgag atgcaccaca gtgcacttgg gtttggaac acattttag gagtagtttt
4860 gttgcaccaa gtgatcatca ggootgattt ggalggcgcc tggcttgaat tttatgggat
4920 agttotagag gccotgcagc agcccacaco atcctgocca ctgcttcacc gataagtcca
4980 gaattcctca gaagtgatct cttgaaagtg ttaaggtttg tggaaaatgt gaattgtttt
5000 tttttttcc octacagaca

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Sequence Number (ID): 11
Length: 4500
Molecule Type: DNA
Features Location/Qualifiers:
- source, 1..4500

Fig. 6-28

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> mol_type. genomic DNA
> organism. Cricetulus griseus
Residues:
60  aacacacato agcatattat tggtgaaato agactgcttg aaatccctaa cattcggtta
120  tgaatcattt tacaatttcc ttgttgcttc tttgttgaat ttaatcccaa agatcttget
180  tottcactta tggaaactta acaccttgga ctcaataago cttcagtggt acccaaggcc
240  tcggaaaact tggtcaggaa ctgcctggag caaaatgcta ggggttaaga gagaaatgca
300  acttcctgta gctgactaga tgatagata accaaagcaa aaaaacatat ttcaatctct
360  ctatgaaaga cgaataggaa aatatagaaa caagagacat tggcttcag acttacagcc
420  cagaagaaac agaaaaatac accggaagag aggtgctgac ttcagccctc ccatgaagaa
480  atatgtgcat tccagagact tgcagattct tgtagtgttt gagacaggat cgtctccctt
540  ggtagcccaa gcttgccctc atatagtagc cagttgcttt ctttcttttt ttattagtat
600  tatatttggg gtttagagat ggaggtcaga ggacagttga cagttctccc tttcagcgct
660  gggtcctagg actcgaactc agggatatcag gcataatggc aggtgccctt gccagagtc
720  atctcactgg ctctaagaca gctgggtttt aatggetatc agtaaccttc tttctttgag
780  atatggtctc atcattttct tgcctggctt ggagctcatt atgtagacaa ggctagcctt
840  gctctcaaga gatctgcctg cttccgtcta cccagccaga gccgtgatta aaccccagat
900  ttactttctt gatggaaaag gttaaatgac ttcacccgcc cttccggtc atttgccca
960  tcgctgtgga atattactag acttctggcc tacataaaac cctaaactca ccgtgctttg
1020  gtggctcaca ctttaatcc cagcactigg gaggcagagg caggcgatct cagttagttc
1080  gaggccagcc ggtctacag taagagttcc aggcagacct ctaaaacat acaaaagaaac
1140  accaaaacaa aaccctaacc tctaaacaca aatcacggcc tcaaaaatgc aggtaggttt
1200  gatgtagttc actaccagag ggaaaagata tattaaatct ttcaagaaag gcaacctaac
1260  ggggaaaagt agtaaatcca cgtgcggtg gagctggggg agaaaaggag gttctaaagg
1320  gtagggtgga gcagctaagt aaagggtgca gcctttactt agcacaagtg ggacctggg
1380  ttgatgccc tgcacacaac aaccasatga ttaggtaatg ttacggggaa gttcaacttc
1440  ctttgctac ctotgcgtga ggtctccgc agcaaggaca aaaggacagg gtcacottgg
1500  cgggtggcac ccgacagcca ggaccaggat ccgcgggggt ggggggggtt gggggtggg
1560  ggcacgcagc gcgcaggcgc ggtgtccac acccccctcc ccgogccgcg ttccacgcga
1620  gtgaggctcc cagctgcgg ottaacgttt ccacgcgcct ggggtctccc acacctcggg
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Fig. 6-29

1680	aagccgagtg	cgtgtgaggac	toggagtacg	tgcggcgggc	eggagcccgc	atgtcggtg
1740	cgtagcgaag	ccccgccctc	aoccagactc	cgcagaccct	cagctcagcc	ccggactagc
1800	aacggggaggc	tcgcccagta	ggccgaaaat	gocacaggcc	cagccaacca	agtctacccc
1860	cgcgcccggg	ccatcgtcct	cagactgicc	gagccaggag	gacaacctag	tcacgtcttt
1920	ctttctggagt	gacagcgccg	ggggccaatg	cccggacgga	gaggacagag	ggcgggcccc
1980	aaacgccaaag	aatgggcggg	ccctcccgag	ctgctcaaaa	agtoggagtg	ggaggcgctt
2040	agaccagctc	gggaaccggc	tcgtgcgcac	gtccctgota	aacgctctcc	agggtgtgta
2100	taggaacctc	tgttcttttc	ctgctcgccc	accggagtgc	caggtaggga	tcgggggccc
2160	ctggggaacg	gttcgctcac	gagctccggc	tggggggccc	cggtaggtgt	gtgtgogccc
2220	ctccagcctg	ggcgcgtgcc	cggggactcg	ggaagcgttc	ggggcagcct	cgtggaggga
2280	cgtgggtctt	tggagggggc	tggggaggtg	cagatccacc	tgtagcatc	cagacacagg
2340	cttgcagggc	aactgggctc	ccaagcctac	aaccgcaagt	gcagaacagg	gtggatcgcc
2400	gccccggggg	gttagctaac	tggagggaag	cttttttttt	tttttttttt	ttgaggggga
2460	gaatgatttg	acotaatgct	caggaaaaac	cttcacaaaa	tggaaggggc	tgaagaagg
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2580	gcagatcctg	ctagggtctg	ggctcatgga	cgcgaatcta	gcgtcacagg	gaggagtg
2640	tcagcccat	ctctgcccag	gaatatctc	cgtgtcttat	tcaggggaa	attgcagcag
2700	aacccctgga	tagctgggga	gggcttaggc	gaactcgccc	cagttaggca	agagttgogg
2760	gcttctgag	tggctgctag	caggccgaga	accgacttag	gatgtgttcc	aggaaatttc
2820	ctgtcgcccc	tggcgtaagc	agcgtggctc	ttttgcttcc	gtgggaggcc	agagccggcc
2880	tggscggata	ctttgacctc	taggtccact	attctgcccc	ggctcaggcc	ttcagtcac
2940	ttaaaagacc	tagagggtga	gatttttctc	ctgggcaaa	acagtgcagc	ccaaacttag
3000	gatccgagtc	agatgcctgc	tggaaaggga	gaaccaactc	ctacacgtta	gcctgacttc
3060	ctagtgcgcg	ctggcacgtg	ggcgtcctcc	cggcccaaat	agatgaaaag	aaaactaaag
3120	ggccattgtg	ccagacaagg	tgatgcgcac	atataaaccc	ggcacagggg	gogctcgac
3180	aggaggatca	cttgagttcg	ggtcaaccog	gtatacatag	gacgtggagg	tgggagaaga
3240	caaagaagta	aatggtatct	aggccagac	tggtatacaa	aagtgtatct	gaagtctaca
3300	tagtaaatca	tgggtgtttt	aagatggagg	ttttttgtaa	catagagggg	taaaaagtag
3360	atacttttct	tgattagggt	gtaaagatcc	tttgtctcac	acaagacaaa	gcaacagaag

Fig. 6-30

```

3420   aagctactoa  gttgaaaceaa aaagatgcct  ttgcagcttc  ggcocttgcc  gtacgagaag
3480   tggagaaccc  agttagataa  agatagttaa  tgaacaataa  tttagttttc  cactcctatc
3540   aattttatgc  taagagtttt  ttgggtaaaa  ttctgtttg  gaaggcctac  tggggaaaaa
3600   gtaagggtat  ttaaacctct  ttagtggtgt  ggccatttct  taaaaatgag  ctaaaatgcc
3660   ctttttgaaa  gaaatatgat  tcagaagcca  tcagaggatc  ttaaataatg  cggtaggat
3720   ggaaccatt  gctgoccaa  tgctaggcaa  gctagcagac  acgtgccagc  agaccagaaa
3780   gtgtagcgt  gatgatcaca  gacgaaggac  ttaogtggtg  ctgccatcca  agcccaattt
3840   ctcaaaacta  ctgaacatgc  ttctgttgcc  atgggtataa  tataacatag  gctgggtggt
3900   tttttgtttt  gttttgtttt  ttgocaaatt  ttacatggtt  tatcctgtga  ctttcacagc
3960   atcatcctac  ttcttttgca  ttaggccttg  agtgaacca  gccattgct  ctaccagtc
4020   ggagaactga  tggttttagt  ttgggtttt  ttatgggttt  tgocctagt  gttacaaaac
4080   aggttttttc  ttctccttag  cactacatgc  taagaacaca  ttgttaaca  tccgcgcgc
4140   tctgggattt  gtatggaagc  cagggtgttg  agcatgccac  ttgagacta  agccatgacc
4200   ttagcctccc  acctgctttt  gtacaaaagc  gtagttttta  ctgatttgag  atgcaccaca
4260   gtgcacttgg  gtttggaac  acattttag  gagtggtttt  gttgcaccaa  gtgatcatca
4320   ggctgattt  gtagggcgc  tggcttgaat  ttatgggat  agttctagag  gccctgcagc
4380   agccacacc  atcctgocca  ctgcttcacc  gataagtcca  gaattcccca  gaagtatct
4440   cttgaaagt  ttaaggttt  tggaatatgt  gaattgtttt  tttttttcc  cctacagaca
4500

```

Sequence Number (ID): 12

Length: 3500

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..3500

> mol_type, genomic DNA

> organism, *Cricetulus griseus*

Residues:

```

60   caggcgatct  cagtgagttc  gaggccagcc  ggtctacag  tacgagttcc  aggacagcct
120  ctaaaacaat  acaaagaaac  accaaaacaa  aaccttaaac  tctaaacaca  aatacaggcc
180  tcaaaaatgc  aggtaggttt  gatgtagttc  actaccagag  ggaaaagata  tattaatctt
240  ttcaagaaag  gcaccttaac  ggggaaaagt  agtaaattca  cgctgcagtg  gagctggggg
300  agaaaggagg  gttctaaagg  gtagggtgga  gcagctaagt  aaagggtgca  gcctttactt
360  agcacaagt  ggacctgga  ttcatgccc  tgcacacaa  aacaaaatga  ttaggtaatg

```

Fig. 6-31

420	ttacggggaa	gttcaotttc	ctttggetac	ctctgcgtga	ggctctccgo	agcaaggaca
480	aaaggacagg	gtcacottgg	cgggtggcac	ccgacagcca	ggaccaggat	ccgcgggggt
540	ggggggggxt	gggggtgggg	ggcaccgagc	gcgcaggcgc	ggtgtccac	acccccctcc
600	ccgcgcgcgc	ttccacgcga	gtgaggctcc	cacgctgcgg	cttaacgttt	ccacgcgcgt
660	ggggtctccc	acacctcggg	gcggggccgg	gcgcggaagg	tcaccaccaa	ggtcagagac
720	cggaactgct	cgcgcctccc	aagccgagtg	cgtgtgggac	tggagtagc	tgcggcgggc
780	cggagcccgc	atgtcggctg	cgtagcgacg	ccccgccctc	acccagactc	cgcagacctt
840	cagctcagcc	cccggctagc	aacgggaggc	tgcgccagta	ggccgaaaat	gccacaggcc
900	cagccaacca	agtctacccc	cgcgcccggg	ccatcgtcct	cagactgtcc	gagccaggag
960	gacacctcag	tcacgctott	cttctggagt	gacagcgccg	ggggccaatg	cccggaccga
1020	gaggacagag	ggcgggcccc	aaacgccaaag	aatgggcggg	ccctcccag	ctgctcaca
1080	agtcggagtg	ggaggcgctt	agaccagctc	gggaacgggc	tctgtgcgac	gtccctgcta
1140	aacgctctcc	agggtgtgta	taggaacctc	tgttttttcc	ctgctcgccc	accggagtg
1200	caggtaggga	tcggggggccc	ctgggggaacg	gttcgctcac	gagctccggc	tggggggccc
1260	cgtgtgggtgt	gtgtgcgccc	ctccagcctg	ggcgcgtgcc	cggggactcg	ggaagcgttc
1320	ggggcagcct	cgtggagggg	cgtgggctct	tggagggggc	tggggaggtg	cagatccacc
1380	tgttagcctc	cagacacagg	cttgcagggc	aactgggctc	ccaagcctac	aaccgcaagt
1440	gcagaacagg	gtggatcgcc	gcccgggggg	gttagctaac	tggaggaagg	cttttttttt
1500	tttttttttt	ttaggggggg	gaatgatttg	acctaatgct	caggaaaaac	cttcacaaaa
1560	tggaaaggcc	tgaagaagg	atgoccaaga	ggcggctcct	cgtcatcgtc	cgcctccccc
1620	cccccccccc	cgggctgctc	gcagatcctg	ctagggtgtg	ggctcatgga	cgcacatcta
1680	gcgtcactgg	gcgggagtg	tcagoccat	ctctgocacg	gaatatactc	cgggtgcttat
1740	tctaggggaa	atgcagcag	aacccctgga	tagctgggga	gggcttaggc	gacctcgccc
1800	cagttaggca	agagttgcgg	gcttctgag	tggctgctag	caggccgaga	accgacttag
1860	gatgtgtttc	aggaaatttc	ctgtcgcccc	tggcgtaagc	agcgttgctc	ttttgctttc
1920	gtgggaggcc	agagccggcc	tgggcgggata	ctttgacctc	taggtccact	attctgcccc
1980	ggctccggcc	ttcagtcac	ttaaaagacc	tagagggtga	gatttttata	ctgggcaaaag
2040	acagtgcagc	ccaacttagg	gacccagctc	agatgcctgc	tggaaaggga	gaaccaactc
2100	ctacacgtta	gcctgacttc	ctagtgcggc	ctggcacgtg	ggcgtccccc	cggoccaa

Fig. 6-32

```

2160  agatgaaaag  aaactaaag  ggccattgtg  ccagacaagg  tgatgogcac  atataaacco
2220  ggcacagggg  gcgtgcgac  aggaggatca  cttgagttcg  ggtcaaccog  gtatacatag
2280  gacgtggggg  tgggagaaga  caaagaagta  aatggtatgt  aggccagac  ttttatcaa
2340  aagtgtatgt  gaagtotaca  tagtaaatca  ttggtgtttt  aagatggagg  tttttgtaa
2400  oatagagggg  taaaaagtag  ataactttct  tgattagggg  gtaaagatcc  tttgtotoac
2460  acaagacaaa  gcaacagaag  aagctactca  gttgaaacaa  aaagatgcct  ttgcagcttc
2520  ggcccttgcc  gtacgagacg  tggagaacco  agttagataa  agatagttaa  tgaacaataa
2580  tttagtttct  caoctatoc  aatttatgtc  taagagtttt  ttggtaaaa  ttctgtottg
2640  gaaggcctac  tggggaaaaa  gtaagggatt  ttaaacctct  ttagtgttgt  ggccattttc
2700  taaaaatgag  ctaaaatgcc  ctttttgaaa  gaaatatgat  tcagaagcca  tcagaggatc
2760  ttaaatatgg  cggtaggat  ggaaccatt  gctgcccacg  tgctaggcaa  gctagcagac
2820  acgtgccagc  agaccagaaa  gtgtagcgct  gatgatoaca  gacgaaggac  ttaoigtgtg
2880  ctgccatcca  agccaatttt  ctcaaaactaa  ctgaacatgc  ttctgttgcc  atgggtataa
2940  tataacatag  gctgggtggt  tttttgtttt  gttttgtttt  ttgctaatt  ttacatggtt
3000  tatcctgtga  ccttcacagc  atcatcctao  ttcctttgca  ttaggccttg  agtgtaacca
3060  gccattgct  ctaccagto  ggagaactga  tggttttagt  ttggggtttt  ttatgggttt
3120  tgccctagtg  gttacaaaac  aggttttttc  ttctccttag  cactacatgc  taagaacaca
3180  tttgttacca  tcccgccgc  tctgggattt  gtatggaagc  cagggtgttg  agcatgccac
3240  ttgagcacta  agocatgacc  ttagcctccc  acotgctttt  gtacaaaagc  gtagttttta
3300  ctgatttgag  atgcaccaca  gtgcacttgg  gtttgggaac  acatttgtag  gagtggtttt
3360  gttgcaccaa  gtgatcatca  ggctgatgtt  ggatggcgcc  tggcttgaat  tttatgggat
3420  agttotagag  gccctgcagc  agcccacacc  atoctgocca  ctgcttcacc  gataagtcca
3480  gaattcctca  gaagtgatct  cttgaaagtg  ttaaggtttg  tggaaaatgt  gaattgtttt
3500  tttttttccc  cctacagaca

```

Sequence Number (ID): 13

Length: 2707

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..2707

> mol_type, genomic DNA

> organism, Cricetulus griseus

Residues:

ggctagcaac gggaggctcg ccagtaggc cgaatggcc acaggccag ccaaccaagt

Fig. 6-33

60	ctacccccgc	ccccgggcca	tcgtcctcag	actgtccgag	ccaggaggac	acctcagtc
120	cgctotttctt	ctggagtga	agcgccgggg	gccaatgccc	ggacggagag	gacagagggc
180	gggcccga	cgcaagaat	ggcgggggcc	tcocgagctg	ctcacaaagt	cggagtggga
240	ggcgottaga	ccagctcggg	aaccggctcg	tgcgcaogtc	cttgcataac	gctctocagg
300	gtgtgtatag	gaacctotgt	tcttttctcg	ctcgccaccc	ggagtgcag	gtaggatcg
360	ggggccctg	gggaacggtt	cgctcagag	ctccgcgtgg	ggggcccg	tgggtgtgtg
420	tgcgcccctc	cagcctgggc	gcgtgcggg	ggactcggga	agcgttcggg	gcagcctcgt
480	ggaggagcgt	gggtcttgg	aggggctgg	ggaggtgcag	atccacctgt	tagcatccag
540	acacaggctt	gcagggcaac	tgggtccca	agcctacaa	cgcaagtgc	gaacagggtg
600	gatcgccgc	cgggggggtt	agctaactgg	aggaaggctt	ttttttttt	ttttttttg
660	agggggagaa	tgatttgacc	taatgtcag	gaaaaacott	cacaaatgg	aagggcotga
720	aagaaggatg	cccaaggagc	ggtccctcgt	catgtccgc	ctcccccgc	cccccccg
780	gctgctcgca	gatcctgcta	ggcctgtggc	tcattgacgc	caatctagcg	tcactgggcg
840	ggagtggctc	agcccatctc	tgccacggaa	tatactcogg	tgcttattct	aggggaaatt
900	gcagcagaac	ccctggatag	ctggggaggg	cttaggcgac	ctcgcccag	ttaggcaaga
960	gttgcgggt	tcctgagtgg	cigtctagcag	gccgagaacc	gacttaggat	gtgtttcagg
1020	aaatttctcg	tgcgccctgg	cgtaagcagc	gtggtccttt	tgttttcgtg	ggaggccaga
1080	gocggcctgg	goggatactt	tgacctctag	gtccactatt	ctgcccgggc	tcgggccttc
1140	agtcactta	aaagacctag	agggtagat	ttttatctcg	ggcaaagaca	gtcagccca
1200	acttagggat	ccgagtcaga	tgcatgtcgg	aaggggagaa	ccaactccta	cacgttagcc
1260	tgacttccta	gtgcgcgtcg	gcacgtgggc	gtcccccgg	cccaaataga	tgaaaagaaa
1320	actaaagggc	cattgtgcca	gacaaggtag	tgccacata	taaaccgggc	acagggggcg
1380	ctgcgacagg	aggatcactt	gagttcgggt	caaccggta	tacataggac	gtgggggtgg
1440	gagaagacaa	agaagtaaat	ggtatttagg	cccagactgt	tatacaaaag	tgtatttgaa
1500	gtctacatag	taaatcattg	gtgttttaag	atggagggtt	tttgtaacat	agagggttaa
1560	aaagtagata	cttttcttga	ttaggggtga	aagatccttt	gtctcacaca	agacaaagca
1620	acagaagaag	ctactcagtt	gaaacaaaaa	gatgcotttg	cagcttcggc	ccttgccgta
1680	cgagacgtgg	agaaccaggt	tagataaaga	tagtttatga	acaataattt	agtttctcac
1740	ctcatccaat	ttattgctaa	gagttttttt	ggtaaaattc	tgtcttgga	ggcctactgg

Fig. 6-34

```

1800      ggaaaaagta  agggatttta  aacctcttta  gtgttggtgc  catttcctaa  aaatgagcta
1860      aaatgccctt  tttgaaagaa  atatgattca  gaagccatca  gaggatctta  aatatggcgg
1920      tgaggatgga  acccattgct  gccacgtgc  taggcaagct  agcagacacg  tgccagcaga
1980      ccagaaagtg  tagcgtgat  gatcacagac  gaaggactta  cgtggtgctg  ccatacaagc
2040      ccaattttct  aaactaactg  aacaigcttc  tgttgccaig  ggtataatat  aacataggot
2100      gggteggttt  ttgttttgtt  ttgttttttg  cctaatttta  catggtttat  cctgtgaocct
2160      tcacagcata  atcctacttc  ctttgcattt  ggccttgagt  gtaaccagcc  cattgtctta
2220      occagtcgga  gaactgatgg  ttttagtttg  gggtttttta  tgggttttgc  cctagtgggt
2280      acaaaacagg  tttttctctc  tcttagcac  tacatgctaa  gaacacattt  gttaccatcc
2340      cgcccgctct  gggatttgta  tgaagccagg  ggtgttgagg  atgccacttg  agcactaagc
2400      catgacotta  gctccacc  tgccttttga  caaaagcgta  gtttttaactg  atttgagatg
2460      caccacagt  cacttggtt  tgggaacaca  ttttaggag  tggttttgtt  gcaccaagtg
2520      atcatcaggc  ctgatttga  tggcgctgg  cttgaatttt  atgggatagt  tctagaggcc
2580      ctgcagcagc  ccacaccatc  ctgccactg  cttcaocgat  aagtcagaa  ttctcagaa
2640      gtgatctctt  gaaagtgtta  aggtttgtgg  aaaatgtgaa  ttgttttttt  ttttccct
2700      acagaca
2707

```

Sequence Number (ID): 14

Length: 2607

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..2607

> mol_type, genomic DNA

> organism, *Cricetulus griseus*

Residues:

```

60      ccaggaggac  acctcagtca  cgctcttctt  ctggagtga  agcgccgggg  gccaatgccc
120      ggacggagag  gacagagggc  gggcccgaaa  cgccaagaat  gggcgggccc  tcccgagctg
180      ctcaaaagt  cggagtggga  ggcgcttaga  ccagctcggg  aaccggctcg  tgcgcacgtc
240      cctgctaaac  gctctccagg  gtgtgtatag  gaacctctgt  tcttttcctg  ctgcccacc
300      ggagtgccag  gtagggatcg  ggggcccctg  ggaacgggtt  cgctcacgag  ctccgcgtgg
360      ggggcccogg  tgggtgtgtg  tgcgcccctc  cagcctgggc  gcgtgcccgg  ggaotcggga
420      agcgttcggg  gcagcctcgt  ggaggagcgt  ggcctcttgg  agggggctgg  ggaggtgcag
480      atcacctgt  tagcatcag  acacaggctt  gcaggccaac  tgggtccca  agcctacaac

```

Fig. 6-35

540	cgcgaagtgc	gaacaggggtg	gacgcgcgc	cggggggggtt	agctaactgg	aggaaggcctt
600	tttttttttt	tttttttttg	agggggagaa	tgatttgacc	taatgctcag	gaaaaacctt
660	cacaaaatgg	aagggcctga	aagaaggatg	cccaagaggo	ggtccctcgt	catcgtcgcg
720	ctcccccccc	ccccccccgg	gotgctcgca	gatcctgcta	gggctgtggc	tcatggaccg
780	caatctagcg	tcactgggcg	ggagtggctc	agcccatctc	tgccacggaa	tatactccgg
840	tgcttattct	aggggaaatt	gcagcagaa	ccctggatag	ctggggaggg	cttaggcgac
900	ctgcccccag	ttaggcgaaga	gttgccggct	tcctgagtgg	ctgctagcag	gccgagaacc
960	gaattaggat	gtgttcagg	aaatttcctg	tcgcccctgg	cgtaagcago	gtggtccttt
1020	tgctttcgtg	ggaggccaga	gcgggcctgg	gcggatactt	tgaacctctag	gtccactatt
1080	ctgccccggc	tcgggccttc	agtcacctta	aaagacctag	agggtagagat	ttttatcctg
1140	ggcaaaagaca	gtgcagccca	acttagggat	ccgagtcaga	tgcattgctgg	aaggggagaa
1200	ccaactccta	cacgttagcc	tgacttccta	gtgcgcgctg	gcacgtgggc	gctccccggg
1260	cccaaataga	tgaaaagaaa	actaaagggc	catgtgccca	gacaaggiga	tgcgcacata
1320	taaaaccggc	acagggggcg	ctgcgacagg	aggatcactt	gagttcgggt	caaccoggta
1380	tacataggac	gtggggatgg	gagaagacaa	agaagtaaat	ggtatttagg	cocagactgt
1440	tatacaaaag	tgtatttgaa	gtctacatag	taaatacattg	gtgttttaag	atggagggtt
1500	tttgtaacat	agaggggtaa	aaagttagata	cttttcttga	ttaggggtga	aagatccttt
1560	gtctcacaca	agacaaaagca	acagaagaag	ctaactcagtt	gaaacaaaaa	gatgcctttg
1620	cagcttcggc	ccttgccgta	cgagacgtgg	agaaccacgt	tagataaaga	tagtttatga
1680	acaataattt	agttctcac	ctcatccaat	ttattgctaa	gagttttttt	ggtaaaattc
1740	tgtcttgga	ggcctactgg	ggaaaaagta	agggatttta	aacctcttta	gtgttgtggc
1800	catttcttaa	aaatgagcta	aaatgccott	tttgaaagaa	atatgattca	gaagccatca
1860	gaggatotta	aatatggcgg	tgaggatgga	accatttgct	gccacgtgco	taggcaagct
1920	agcagacacg	tgccagcaga	ccagaagtg	tagcgctgat	gatcacagac	gaaggactta
1980	cgttgtgctg	ccatccaagc	ccaatttctc	aaactaactg	aacatgotto	tgttgocatg
2040	ggtataatat	aacataggct	gggtggtttt	ttgttttggt	ttgttttttg	octaatttta
2100	catgttttat	cotgtgacct	tcacagcatc	atcctacttc	cttfgcatta	ggccttaggt
2160	gtaaccagcc	cattgotcta	ccagctogga	gaaotgatgg	ttttagtttg	gggtttttta
2220	tgggttttgc	cotagtgttt	acaaaacagg	tttttcttc	tccttagcac	tacatgotaa

Fig. 6-36

```

2280   gaacacattt  gtiaccatcc  cgcccgcctc  gggatttgta  tggaaagcoag  ggtgttgagc
2340   atgccacttg  agcaactaagc  catgacctta  gcctcccacc  tgcitttgta  caaaagcgta
2400   gtttttactg  atttgagatg  caccacagtg  cacttgggtt  tgggaacaca  ttgtaggag
2460   tggttttgtt  gcaccaagtg  atcatcaggo  ctgatttgga  tggcgccctg  cttgaatttt
2520   atgggatagt  totagaggcc  ctgcagcago  ccacacccatc  ctgcccactg  cttcacogat
2580   aagtcagaa  ttctcagaa  gtgatctctt  gaaagtgtta  aggtttgttg  aaaatgtgaa
2607   ttgttttttt  ttttccct  acagaca

```

Sequence Number (ID): 15

Length: 2507

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..2507

> mol_type, genomic DNA

> organism, *Cricetulus griseus*

Residues:

```

60   gggcgggccc  tccgagctg  ctcacaaagt  cggagtggga  ggcgottaga  ccagcteggg
120  aaccggctcg  tgcgcaogtc  cctgetaaac  gctctccagg  gtgtgtatag  gaacctctgt
180  tottttctcg  ctgcccacc  ggagtgcag  gtagggatcg  ggggcccctg  gggaacgggt
240  cgctcacgag  ctccgctgg  ggggcccgg  tgggtgtgtg  tgcgcccctc  cagcctgggc
300  gcgtgcccgg  ggactcggga  agcgttcggg  gcagcctcgt  ggaggagcgt  gggctcttgg
360  agggggctgg  ggaggtgcag  atccacctgt  tagcatccag  acacaggctt  gcaggccaac
420  tgggctocca  agcctacaac  cgcaagtcca  gaacagggtg  gatcgccgco  cgggggggtt
480  agctaactgg  aggaaggctt  tttttttttt  tttttttttg  agggggagaa  tgatttgacc
540  taatgctcag  gaaaaacctt  cacaaaatgg  aagggcctga  aagaaggatg  cccaaggaggc
600  ggtccctcgt  catcgctcgc  ctcccccctc  cccccccgg  gctgctcgca  gatcctgcta
660  gggctgtggc  tcatggaccg  caatctagcg  tcaotgggcg  ggagtggtoc  agcccatctc
720  tgccacggaa  tatactcogg  tgcttattct  aggggaaatt  gcagcagaac  ccctggatag
780  ctggggaggg  cttaggogac  ctgcccacag  ttaggcaaga  gttgcgggct  tctgagtg
840  ctgctagcag  gcgagaacc  gacttaggat  gtgttcagg  aaatttctcg  tcgcccctgg
900  cgtaagcagc  gtggtccttt  tgctttcgtg  gggggcaga  gccggcctgg  gggataactt
960  tgacctctag  gtccactatt  ctgcccgggc  tccggccttc  agtccactta  aaagacctag
1020  agggtgagat  ttttatctcg  ggcaagaca  gtgcagcca  acttagggat  ccgagtcaga
      tgcattctgg  aaggggagaa  ccaactccta  cacgttagcc  tgacttcccta  gtgcgcctg

```

Fig. 6-37

```

1080      gcacgtgggc  gtcoccccg  cccaaataga  tgaaaagaaa  actaaagggc  catttgcca
1140      gacaaggatga  tgcgcacata  taaaccocgg  acagggggcg  ctgcgcacagg  aggatcaatt
1200      gagttcgggt  caaccocggt  tacataggac  gtgggggtgg  gagaagacaa  agaagtaa
1260      ggtatttagg  ccagactgt  tatacaaaag  tgtatttgaa  gtctacatag  taaatcattg
1320      gtgttttaag  atggagggtt  ttgttaacat  agaggggtaa  aaagtagata  cttttcttga
1380      ttaggggtga  aagatccttt  gtctcacaca  agacaaagca  acagaagaag  ctactcagtt
1440      gaaacaaaaa  gatgcctttg  cagcttcggc  ccttgccgta  cgagacgtgg  agaaccagt
1500      tagataaaga  tagtttatga  acaataattt  agttctctac  ctcatccaat  ttatigctaa
1560      gagttttttt  ggtaaaatto  tgtcttgtaa  ggctaactgg  ggaaaaagta  agggatttta
1620      aacctcttta  gtgtgtggc  catttcttaa  aaatgagcta  aaatgccctt  ttigaagaa
1680      atatgattca  gaagccatca  gaggatctta  aatatggogg  tgaggatgga  acccattgct
1740      gccacgtgc  taggcaagct  agcagacacg  tgccagcaga  ccagaaagtg  tagcgtgat
1800      gatcacagac  gaaggactta  cgtggtgtg  ccaccaagc  ccaatttctc  aaactaactg
1860      aacatgcttc  tgttgccatg  ggtataatat  aacataggct  ggggtggttt  ttgtttgtt
1920      ttgttttttg  cctaatttta  catggtttat  cctgtgacct  tcacagcatc  atcctacttc
1980      ctttgcatta  ggcottgagt  gtaacagac  cattgctcta  ccagtcgga  gaactgatgg
2040      ttttagtttg  gggtttttta  tgggttttgc  cctagtgggt  acaaaacagg  tttttcttc
2100      tcottagcac  tacatgctaa  gaacacattt  gttaccatcc  cggccgctct  gggatttgta
2160      tggaagccag  ggtgttgagc  atgccacttg  agcactaagc  catgaactta  gctcccaacc
2220      tgottttgta  caaaagcgta  gtttttactg  attgagatg  caccacagtg  cacttgggtt
2280      tgggaacaca  ttgtaggag  tggttttgtt  gcaccaagtg  atcatcaggc  ctgatttgga
2340      tggcgootgg  cttgaatttt  atgggatagt  tctagaggcc  ctgcagcagc  ccacaccatc
2400      ctgccactg  ctccaccgat  aagtccagaa  ttctcagaa  gtgatctctt  gaaagtgtta
2460      aggtttgtgg  aaaaatgtaa  ttgttttttt  ttttccct  acagaca
2507

```

Sequence Number (ID): 16

Length: 4000

Molecule Type: DNA

Features Location/Qualifiers:

```

- source, 1..4000
  > mol_type, genomic DNA
  > organism, Mus musculus

```

Residues:

Fig. 6-38

60	taactoccag	actcacaacc	caagagaaac	gtgggagaaa	atacaacaga	aagtgatgtg
120	ttccottcag	ttctatagtg	aagtgtgtgc	acggcaggag	ttgaaagccg	atttctgtag
180	tgtttgagat	gagatcgggt	ccctaggcag	ctcaagtttg	tctcgatttg	atggctactt
240	tttttctttt	gattagtatt	acacattgtg	gttgggatg	gaggtcagag	gacagcttga
300	tcaggtgtga	ctgactgoag	gtgcctttgc	ccctgagoca	tctcgtgggt	tccaagatag
360	ggttttgttt	gtcgtcttta	ttttaaaata	ttgtttgggt	ttgggtttgt	ttgtttgggt
420	ttttttggac	agggtttctc	cttgtaactt	atctgttcag	gaactagctc	tgtagattaa
480	aggtccacat	tagcaactgac	tggttagcta	tcctcacctt	tattttttct	ctttcttcca
540	tttctttctt	tttctctttt	cttttctttt	ctttctttct	tttttctttt	ctttctttct
600	ttctctttct	ttttttttt	tttttgacaa	ggtctcatca	ctatgcccc	tctgaattag
660	aattcactat	gtagaccagg	caggccctga	actcagaaat	cgccttgcc	ctgcctaccc
720	acagctgtgg	ttaaacacca	gctttatttt	cttcctcaga	agggtgaaat	ggctccaccc
780	tgcgcttccg	gacgtttgtc	acatcatigt	ggaatattac	tagtctagac	tttgagaaa
840	tgogtcggga	cgacttgagt	atgagtctca	agaactggta	ggtcagaac	atcaggcca
900	caagcagggg	aaagaagtat	tagatcttta	aacaaaggaa	cattaacatg	gatgaagaat
960	aaatccacac	tgagctagta	caocagagaa	gtaaggaaag	ttctaaagta	gggtggagtg
1020	gctgggtggg	tgiagcacia	aggtactgcc	ttattagcgt	gagctggaca	gtaggattcg
1080	gtaccccca	cacagtaaca	aattgattagg	taataataag	ggaaaggtca	cttcccttg
1140	gctacctcgg	ccgagactc	actacagcaa	ggacaaaagg	acggggtcac	cttggcagat
1200	ggcaccagag	agccaggacc	agcggcgagg	gcgtcagtg	cgcaggcgca	gtatccatgc
1260	atttcccoct	catccccccc	accccccccc	aattccacgc	gggagtgagg	ccctctggcc
1320	gcgcatcaac	gittccacac	cgccgggggc	tccagaacct	cccggtggg	ggttgccgg
1380	ctggagggcc	gagccgcgaa	ggtcatcagc	aaggctgttg	gcgggacgig	tggcgccgca
1440	gccacgccga	gtcgtgtga	gactcggagt	acgtgcggcg	ggctcaagcc	ctcatgcccg
1500	cgcgcagcc	cagccccgcc	ccgccccgca	tccgggctcc	gtagaccctc	ggccggggcc
1560	caggttccca	gcccggaggct	cgccaggcat	tccagctatg	ccccaaagag	gtaggccca
1620	gaaaggcagg	acaacccccg	cccccgctgc	ctcgcccta	gtctgggcag	gcgagcgcc
1680	cgcctcggtc	acgtctttct	attggagtga	tggcagctg	agccaatggg	gaggcgccgc
1740	ggaggggggg	cgggcccgaa	aggggggaac	tggcggggc	ttacagagca	ggaaaggag

Fig. 6-39

1800	acagagtgag	aggcgcttag	tgtgtgtccg	gtacagggtc	gcgctogcgt	ctgtccttaa
1860	ggctctcctc	ggtgtccacg	gcacctcttt	tccttgcttt	gcagcgatcc	tactgccagg
1920	tatgggtcctt	gtctccaccg	gggtcagctc	tcctactagc	ttcgcttggg	gaccgggtgg
1980	gcacgtgcac	ccactagcct	ggcggggagc	cgctatccgg	ggagcactcg	ggtcagcctc
2040	gtggaaggac	gcaggctctt	ggagggtgga	ggtgtgagcc	ccaccagtga	tcaccgagtt
2100	tcagggttgc	aatgtaaccg	ggttcccaag	cagacaacag	caaatgcaga	ccacggatga
2160	tgcgggccgt	ggagaataga	aaagtacttt	ctgtgtagtt	aagtgggaagg	tttcagaggg
2220	agggagagca	acttaaaccc	ccgatgtctc	ttaaatcctt	tctcagaatg	gaagggccag
2280	aaataaggat	gccaagagg	cagagcgcca	cacctagta	gatcctgctg	gggctgtagc
2340	tcctggaccg	caatcgagcg	tcactgggcg	ggagtggctc	agcccatctc	tgtgtggaa
2400	tattcctgac	ctgtgagtgt	cggttaggta	ctaagagtag	aggaaattgc	agcagtcccg
2460	ttggataatt	gggaagccct	caggagacct	cgaogcaatt	aggcaagagc	tacaggttcc
2520	ctgggtggct	actagccggc	cgtgaactga	cttaagatgc	atttaagaaa	acttctgttt
2580	ogttctggcc	taggoagtgt	ggttccctgg	ctgactogct	ttogggatag	accacagcta
2640	gtctgtgat	tagcctgaac	gataggcca	ctttctgccc	cttgggtctg	cgttcaatcc
2700	agataaagac	agagtgggtg	agattttctt	cctttggtta	aatacgtttg	ctgcctagcc
2760	taaagatcca	agctagataa	aaagggtgga	gggaggatcc	aactcctaca	agttgtaagg
2820	tccttgcaga	gcaaagtgtg	gcacgtgggt	gtctctccgc	cccataaag	gaaccagcca
2880	tggagatgca	cacatgaaag	tgtagcgcta	gggatgctga	ggctggagga	tcactgagtt
2940	ctggtcagtc	tggcattcat	agaatcgggc	gagtaagaga	gaagacaaaa	gagtatggtt
3000	gaggaatagt	aaggttttgt	aacatggtag	gtaaaaagta	gatactttat	tcttgggtgag
3060	tgtacaaaga	tcctttgtct	tagagaagac	aaaaccacag	aagtaacttc	tgtggaaaaa
3120	aaagctgcct	tigcttcagt	gcttgccaca	aaagatgtga	ggaacctagt	tagataataa
3180	tagcctctga	atgctgtatt	gtcttcattgt	ctcatccagt	ttagctacag	aatctatttg
3240	gtttctttct	tggaaagcct	gctggggaag	aagtaagaaa	gggaacttga	gtttgttgcc
3300	aaatatcctc	ttcctctcag	gatgtttttc	taattaaaca	totactttga	atacctttta
3360	aaagcgagac	aagggtgccct	tttgaaatgt	actaaccaga	agtcctagct	cttgccataa
3420	tatggcggag	attgaaccca	ggacctccca	cgtgctagcc	agcatggcca	ccaggccaag
3480	aatggcagga	tcattgacca	caggccaagg	agtttggggt	tctggcctcc	tgcctcacc

Fig. 6-40

```

3540   ttcttactg gttatcaaat ctgigtacag tgaatatct gctcttgta tgggtgcaat
3600   ttgatgaatt agggtttctc tttttccac aattaaactg ctttgaaatg tcacatttcc
3660   ctctgtgtct caggtcagtt taacctctga ccttccttca ctgttgcttc ccacttcott
3720   ctctgtggag ccagaacact gttctatgca gtcagaggat ttaatttgcc cacagtacag
3780   gtagcttctg agggctttct gtaggagtag tttatttcat tgtgccatgt tgacctcaag
3840   cttggcggga agtccaagcc aaccacctgc tgtgtaagga gtagtcctcg ctccccaacc
3900   cctcactgat gagtgaactt cctgtaigtg aactaacgca tagccatttt ggtaacctgc
3960   aagtcagga cccctcagag gtgatggcct tggaaatgaat taacctctgt gtcactgttt
4000   attggtgata tgaatgtttt tctcctaca gaaattgccc

```

Sequence Number (ID): 17

Length: 3500

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..3500

> mol_type, genomic DNA

> organism, Mus musculus

Residues:

```

60   cttttctttt cttttctttt ttcttttttt cttttctttt ttcttttttt tttttttttt
120   tttttgacaa ggtctcatca ctatgcccc aatgaattag aattcactat gtagaccagg
180   caggccttga actcagaaat ccgcctgcct ctgcctaccc acagctgtgg ttaaaccacca
240   gctttatttt cttoatcaga agggtgaaat ggctccaccc tgcgcttcgg gacgtttgtc
300   acatcattgt ggaatattac tagctagac tttgagaaa tgcgtcggga cgacttgagt
360   atgagtctca ageactggtg ggtcaagaac atcgaggcca caagcagggg aaagaagtat
420   tagatcttta aacaaaggaa cattaacatg gatgaagaat aaatccacac tgagctagta
480   caccagagaa gtaaggaaag ttctaaagta ggttgagtg gctgggtggg tgtagcacia
540   aggtactgcc ttattagcgt gagctggaca gtaggattcg gtacctcca cacagtaaca
600   aatgattagg taatattaag ggaagggtca ctttccttg gctacctcgg cctgagactc
660   actacagcaa ggacaaaagg acggagtcac ctgggcagat ggcaccagag agccaggacc
720   agcggcgagg gcgctcagtg cgcaggcgca gtatccatgc atttccccct catccccccc
780   acccccaccc aattccacgc gggagtggg cctcttgcc gcgcataaac gtttccacac
840   cgcggggggc tccagaaact cccggtgggg ggttggccgg ctggaggggc gagcggcgaa
900   ggtcatcagc aaggtcgtgg gcgggacgtg tggcgccgca gccacgccga gtcggtgtga
gactcggagt acgtcgggg ggctcaagcc ctcatgcccg ccgcgcagcc cagccccggc

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Fig. 6-41

960	cgcgcgcgca	tccgggctcc	gtagaccctc	ggccggggcc	cagggttcca	gcccggaggt
1020	cgccaggcat	tccagctatg	ccccaagcag	gctaggccca	gaaaggcagg	acaacccccg
1080	cccccgctcc	ctcgcacctc	gtctgggcag	gccgagcgcc	cgccctcggtc	acgtctctct
1140	attggagtga	tggcaagctg	agccaatggg	gaggcggcgc	ggaggaggag	cgggcccgaa
1200	aggggagAAC	tgggcggggc	ttaacgagca	ggaaaggag	acagagtggg	aggcgcttag
1260	tgctgctcgg	gtacagggtc	gogctcgggt	ctgtccttaa	ggctctcttc	ggtgtccacg
1320	gcacctcttt	tccttgcttt	gcagcgatcc	tactgccagg	tatggctctt	gtctccaccg
1380	gggtcagctc	tcctactagc	ttcgcttggg	gaacccggtg	gcacgtgcac	ccactagcct
1440	gggcgggagc	cgtatccgg	ggagcactcg	ggtcagcctc	gtggaaggac	gcaggctctt
1500	ggagggtgga	ggtgtgagcc	ccaaccagta	tcaccgagtt	tcagggttgc	aatgtaaccg
1560	ggttcccaag	cagacaacag	caaatgcaga	ccacggatga	tcgcccgggt	ggagaataga
1620	aaagtacttt	ctgtgtagtt	aagtggaaagg	tttcagaggg	agggagagca	acttaaaccc
1680	ccgatgctca	ttaaatcctt	tctcagaatg	gaaggggcag	aaataaggat	gccaagagg
1740	cagagcgoca	caccctagta	gatcctgctg	gggctgtagc	tcctggaccg	caatcgagcg
1800	tcactgggcg	ggagtgggtc	agcccatctc	tgctgtggaa	tattcctgac	ctgtgagtgt
1860	cggtaggta	ctaagagtag	aggaaattgc	agcagtcccg	tggataaatt	gggaagccct
1920	caggagacct	cgacgcaatt	aggcaagagc	tacaggtttc	ctgggtgggt	actagccggc
1980	cgtgaactga	cttaagatgc	atttaagaaa	acttccgtgt	cgttctggcc	taggcagtgt
2040	ggttcccttg	ctgactcggt	ttcgggatag	accacagcta	gtctgtgat	tagcctgacc
2100	gatagggtcc	ctttcttgcc	cttggctotg	cgttcaatcc	agataaagac	agagtgggtg
2160	agattttctt	cttttggtta	aatacgtttg	ctgcctagcc	taaagatcca	agctagataa
2220	aaagggtgga	gggagggtc	aactcctaca	agttgtaagg	tctttgcaga	gcaaagtgtg
2280	gcacgtgggt	gctcctccgc	ccccataaag	gaaccagcca	tggagatgca	cacatgaaag
2340	tgtagcgcta	gggatgctga	ggctggagga	tcactgagtt	ctggtcagtc	tggcattcat
2400	agaatcgggc	gagtaagaga	gaagacaaaa	gagtatggtt	gaggaaatag	aaggttttgt
2460	aacatggtag	gtaaaaagta	gatactttat	tcttgggtgag	tgtacaaaga	tcctttgtct
2520	tagagaagac	aaaaccacag	aagtaacttc	tgtggaaaaa	aaagctgcct	ttgcttcagt
2580	gcttgccaca	aaagatgtga	ggaacctagt	tagataataa	tagcctctga	atgtgttatt
2640	gtcttcatgt	ctcatocagt	ttagctacag	aatctatttg	gtttctttct	tggaaagcct

Fig. 6-42

```

2700      gotggggaag aagtaagaaa gggacttgca gtttggtgco aaatatctctc ttctctctcag
2760      gatgtttttc taattaaaca totactttga atacctttta aaagcgagac aagggtgcctt
2820      tttgaaatgt actaaccaga agtcttagct ottgcctaaa tatggcggag attgaaccca
2880      ggacctccca cgtgctagcc agcatggcca ccaggccaag aatggcagga tcattgaacca
2940      caggccaagg agtttggggt totggcctcc tgcctcacac ttctttactg gttatcaaat
3000      ctgtgtcagc tgaatattct gctcttgtoa tgggtgcaat ttgatgaatt agggtttctc
3060      tttttccac aattaaactg otttgaatg tcacatttcc ctotggtctc caggtcagtt
3120      taacctctga ccttcttca ctgttgcttc ccacttctt ctotgtggag ccagaacact
3180      gttctatgca gtcagaggat ttaatttgcc cacagtacag gtagcttctg agggctttct
3240      gtaggagtgg tttatttcat tgtgccatgt tgacctcaag ottggcggga agtccaagcc
3300      acgacotgc tgtgtaagga gtagtctctg ctccocaaac cctcactgat gagtgaacttt
3360      cctgtatggt acctaacgca tagcccatgt ggtacctgtc aagtcagga cccctcagag
3420      gtgatggcct tggaatgaat taacctctgt gtcactgttt attggtgata tgaatgtttt
3480      tctctctaca gaaattogcc
3500

```

Sequence Number (ID): 18

Length: 100

Molecule Type: DNA

Features Location/Qualifiers:

- source, 1..100

> mol_type, genomic DNA

> organism, *Cricetulus griseus*

Residues:

```

60      gtgggactcg gagtacgtgc ggcgggcccg agcccgcctg toggctgcgt agcgacgccc
100     cggcctcacc cagaotccgc agacotcag ctcagccccc
END

```

PROMOTER OF ENO1 GENE**TECHNICAL FIELD**

[0001] The present invention relates to transfected mammalian cells with enhanced transcriptional activity in relation to a foreign protein, which is obtained by using a foreign gene expression vector having an Eno1 gene promoter, and a method for producing the foreign protein using the same.

BACKGROUND ART

[0002] The development of genetic engineering techniques has rapidly expanded the market of protein-based pharmaceutical products such as therapeutic proteins and antibody drugs. Among them, antibody drugs are considered to have little risk of causing adverse immune responses when administered to the human body and are under active development because of their high specificity.

[0003] Examples of hosts to produce the protein-based pharmaceutical products typified by antibody drugs can include microorganisms, yeasts, animal and plant cells, insects, and transgenic animals and plants. Posttranslational modification such as correct folding or sugar chain modification is often essential for the physiological activity or antigenicity of the protein-based pharmaceutical products. Therefore, microorganisms which cannot perform sugar modification, or plants which differ significantly in sugar chain structure from humans are not suitable as hosts. Cultured mammalian cells, such as CHO (Chinese hamster ovary) cells are currently mainstream, which can produce the protein-based pharmaceutical products possessing a sugar chain structure similar to that of humans, permit posttranslational modification, and are confirmed to cause no safety concern from their many usage records.

[0004] Use of cultured mammalian cells as a host presents problems such as low growth rates, low productivity and high cost as compared with microorganisms or the like (Non Patent Literature 1). Furthermore, clinical utilization of antibody pharmaceutical products requires administering the pharmaceutical products at large doses. Therefore, a lack of sufficient production capacity thereof has been a global issue. In the case of producing a protein-based pharmaceutical product in a cultured mammalian cell expression system, reduction in production cost has been attempted by making improvements to each production step, because the production cost is higher than that of synthetic low-molecular weight pharmaceutical products. However, increasing the amount of protein produced in the cultured mammalian cell expression system is also a promising method for reduction in production cost (Non Patent Literatures 2 and 3). Accordingly, in order to increase the productivity of a foreign protein in cultured mammalian cells, many approaches such as promoters, enhancers, drug selection markers, gene amplification and culture engineering approaches have been practiced so far through trial and error.

[0005] In the case of using CHO cells as host cells, a human cytomegalovirus major immediate early promoter (hereinafter, referred to as a CMV promoter) derived from a virus is generally used for the expression of foreign genes, i.e., the production of protein-based pharmaceutical products (Non Patent Literatures 4, 5 and 6). It is also known that a polynucleotide upstream of the transcription start site of the gene (promoter region) of elongation factor-1 alpha (EF1 α) (Patent Literature 1 and Non Patent Literature 7) or

the human ribosomal protein RPL32 or RPS11 can be used alone or in combination with an additional heterologous promoter in protein expression in CHO cells (Non Patent Literature 8 and Patent Literatures 2 and 3). The heat-shock protein A5 (Hspa5/GRP78) gene promoter and the heat shock protein A8 (Hspa8) promoter are known to improve the productivity of a foreign protein in mammalian cells (Patent Literatures 4 and 5).

CITATION LIST**Patent Literature**

- [0006]** Patent Literature 1: Japanese Patent No. 3051411
- [0007]** Patent Literature 2: International Publication No. WO 2006/123097
- [0008]** Patent Literature 3: International Publication No. 2013/080934
- [0009]** Patent Literature 4: International Publication No. 2018/066492
- [0010]** Patent Literature 5: International Publication No. 2020/032153

Non Patent Literature

- [0011]** Non Patent Literature 1: Florian M. Wurm., Nat. Biotechnol. 22 (11): 1393-1398, 2004
- [0012]** Non Patent Literature 2: Farid S S., J Chromatogr B Analyt Technol Biomed Life Sci. 848 (1): 8-18, 2007
- [0013]** Non Patent Literature 3: Werner R G. Economic aspects of commercial manufacture of biopharmaceuticals. J Biotechnol. 113 (1-3): 171-182, 2004
- [0014]** Non Patent Literature 4: Durocher Y et al., Curr Opin Biotechnol. 20 (6): 700-707, 2009
- [0015]** Non Patent Literature 5: Boshart M et al., Cell. 41 (2): 521-530, 1985
- [0016]** Non Patent Literature 6: Foecking M K et al., Gene. 45 (1): 101-105, 1986
- [0017]** Non Patent Literature 7: Deer J R. and Allison D S., Biotechnol. Prog. 20:880-889, 2004
- [0018]** Non Patent Literature 8: Hoeksema F. et al., Biotechnology Research International, Volume 2011, Article ID 492875, 11 pages

SUMMARY OF INVENTION**Technical Problem**

[0019] An object of the present invention is to provide a promoter having high foreign gene expression-enhancing activity in host cells such as cultured mammalian cells, and to provide an approach to enhancing the amount of a foreign protein, which serves as a protein-based pharmaceutical product, produced using the promoter.

Solution to Problem

[0020] For solving the above-described problems, the present invention provides the following [1] to [34].

- [0021]** [1] A polynucleotide that is a Chinese hamster-derived gene promoter, the polynucleotide comprising a nucleotide sequence that is selected from SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13 and SEQ ID NO: 14, or has 85% or higher sequence

- identity to each of the sequences, or a partial sequence of the nucleotide sequence.
- [0022] [2] The polynucleotide according to [1], which comprises a nucleotide sequence that is selected from SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11 and SEQ ID NO: 12, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.
- [0023] [2a] The polynucleotide according to [1] or [2], which comprises a nucleotide sequence of SEQ ID NO: 18, or a nucleotide sequence having 85% or higher sequence identity to the sequence.
- [0024] [3] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 1.
- [0025] [4] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 3.
- [0026] [5] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 4.
- [0027] [6] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 10.
- [0028] [7] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 11.
- [0029] [8] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 12.
- [0030] [9] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 13.
- [0031] [10] The polynucleotide of according to [1], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 14.
- [0032] [10a] A polynucleotide that is a Chinese hamster-derived gene promoter, the polynucleotide comprising a nucleotide sequence of SEQ ID NO: 18, or a nucleotide sequence having 85% or higher sequence identity to the sequence.
- [0033] [10b] The polynucleotide according to [10a], which comprises a nucleotide sequence that is selected from SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12 and SEQ ID NO: 13, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.
- [0034] [11] A polynucleotide that is a mouse-derived gene promoter, the polynucleotide comprising a nucleotide sequence that is selected from SEQ ID NO: 2, SEQ ID NO: 17 and SEQ ID NO: 16, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.
- [0035] [11a] The polynucleotide according to [11], which comprises a nucleotide sequence having 85% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 18.
- [0036] [12] The polynucleotide according to [11], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 2.
- [0037] [13] The polynucleotide according to [11], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 17.
- [0038] [14] The polynucleotide according to [11], which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 16.
- [0039] [14a] A polynucleotide that is a mouse-derived gene promoter, the polynucleotide comprising a nucleotide sequence having 85% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 18.
- [0040] [14b] The polynucleotide according to [14a], which comprises a nucleotide sequence that is selected from SEQ ID NO: 2, SEQ ID NO: 17 and SEQ ID NO: 16, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.
- [0041] [15] A polynucleotide comprising a nucleotide sequence having 90% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 2, SEQ ID NO: 17 or SEQ ID NO: 16 in [1] or [11], or a partial sequence of the nucleotide sequence, the polynucleotide having promoter activity.
- [0042] [16] A polynucleotide comprising a nucleotide sequence having 95% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 2, SEQ ID NO: 17 or SEQ ID NO: 16 in [1] or [11], or a partial sequence of the nucleotide sequence, the polynucleotide having promoter activity.
- [0043] [17] A polynucleotide that hybridizes under stringent conditions to a polynucleotide consisting of a nucleotide sequence complementary to the nucleotide sequence of SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 2, SEQ ID NO: 17 or SEQ ID NO: 16 in [1] or [11], the polynucleotide having promoter activity.
- [0044] [17a] The polynucleotide according to any one of [1] to [17], which has a length of 1 kbp or more and 10 kbp or less, 2 kbp or more and 6 kbp or less, 2.6 kbp or more and 5 kbp or less, 2.7 kbp or more and 4.5 kbp or less, or 2.8 kbp or more and 4 kbp or less.
- [0045] [18] A polynucleotide comprising the polynucleotide according to any one of [1] to [17] and [17a] and a foreign gene.
- [0046] [19] A foreign gene expression unit comprising the polynucleotide according to any one of [1] to [18] and [17a] and a foreign gene.
- [0047] [19a] The foreign gene expression unit according to [19], which further comprises a transcriptional terminator region.

- [0048] [20] The foreign gene expression unit according to [19] or [19a], wherein the foreign gene is a gene encoding a protein.
- [0049] [21] The foreign gene expression unit according to [20], wherein the foreign gene is a gene encoding a heteromultimeric protein.
- [0050] [22] The foreign gene expression unit according to [21], wherein the foreign gene is a gene encoding an antibody or an antigen-binding fragment thereof.
- [0051] [23] A foreign gene expression vector comprising the foreign gene expression unit according to any one of [22] and [19a].
- [0052] [24] A foreign gene expression vector comprising the foreign gene expression unit according to any one of [22] and [19a], and one or more polynucleotides selected from the following (a) to (e):
- [0053] (a) a polynucleotide comprising the nucleotide sequence of SEQ ID NO: 5, or a partial sequence thereof;
- [0054] (b) a polynucleotide comprising the nucleotide sequence of SEQ ID NO: 6, or a partial sequence thereof;
- [0055] (c) a polynucleotide comprising the nucleotide sequence of SEQ ID NO: 7, or a partial sequence thereof;
- [0056] (d) a polynucleotide comprising a nucleotide sequence having 80% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 5, SEQ ID NO: 6 or SEQ ID NO: 7, or a partial sequence of the nucleotide sequence, the polynucleotide having foreign gene expression-enhancing activity; and
- [0057] (e) a polynucleotide comprising a nucleotide sequence having 85% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 5, SEQ ID NO: 6 or SEQ ID NO: 7, or a partial sequence of the nucleotide sequence, the polynucleotide having foreign gene expression-enhancing activity.
- [0058] [24a] A foreign gene expression vector comprising the foreign gene expression unit according to any one of [22] and [19a], and the following polynucleotide (b) or (d):
- [0059] (b) a polynucleotide comprising the nucleotide sequence of SEQ ID NO: 6, or a partial sequence thereof; or
- [0060] (d) a polynucleotide comprising a nucleotide sequence having 80% or higher identity to the nucleotide sequence of SEQ ID NO: 6, or a partial sequence of the nucleotide sequence, the polynucleotide having foreign gene expression-enhancing activity.
- [0061] [25] A transfected cell into which the foreign gene expression vector according to [23], or [24a] has been introduced.
- [0062] [26] A transfected cell according to, wherein the cell is a cultured cell derived from a mammal.
- [0063] [27] The transfected cell according to [25], wherein the cell is a COS-1 cell, a 293 cell, or a CHO cell.
- [0064] [28] A method for producing a foreign gene-derived protein, comprising culturing the transfected cell according to any one of [25] to [27], and obtaining the foreign gene-derived protein from the culture.
- [0065] [29] The production method according to [28], wherein the foreign gene-derived protein is a monomeric protein or a multimeric protein.

- [0066] [30] The production method according to [29], wherein the multimeric protein is a heteromultimeric protein.
- [0067] [31] The production method according to [30], wherein the heteromultimeric protein is an antibody protein.
- [0068] [31a] The production method according to any one of [28] to [31], wherein the culture of the transfected cell is fed-batch culture.
- [0069] [31b] The production method according to any one of [28] to [31] and [31a], wherein the transfected cell is cultured for 6 or more days.
- [0070] [32] Use of the polynucleotide according to any one of [1] to [18] for expressing a foreign gene in a transfected cell.
- [0071] [33] Use of the foreign gene expression vector according to [23] or [24] for the purpose of expressing a foreign gene in a transfected cell.
- [0072] [34] A method for producing a transfected cell which expresses a foreign gene, the method comprising performing transfection with the foreign gene vector according to [23], or [24a].

Advantageous Effects of the Invention

[0073] The present invention provides a promoter having high foreign gene expression-enhancing activity in host cells such as cultured mammalian cells, and an approach to enhancing the amount of a foreign protein, which serves as a protein-based pharmaceutical product, produced using the promoter.

BRIEF DESCRIPTION OF DRAWINGS

[0074] FIG. 1 schematically shows humanized antibody gene Y expression vectors pDSLH3.1-Eno1-Y and pDSLH3.1-hEF1 α -Y containing an Eno1 gene or human EF10 gene-derived promoter as a promoter for antibody H chain (HC) and L chain (LC) gene expression.

[0075] FIG. 2-A The amount of antibody produced in fed-batch culture using a humanized antibody Y-expressing stable pool was compared between expression under a Chinese hamster-derived Eno1 gene promoter and expression under a human EF1- α gene promoter. In FIG. 2-A, the vertical axis represents the number of viable cells on each day of sampling. hEF1 α denotes a human EF1 α gene promoter, and Eno1 denotes a Chinese hamster-derived Eno1 gene promoter (the same applies to FIG. 2-B).

[0076] FIG. 2-B The amount of antibody produced in fed-batch culture using a humanized antibody Y-expressing stable pool was compared between expression under a Chinese hamster-derived Eno1 gene promoter and expression under a human EF1- α gene promoter. In FIG. 2-B, the vertical axis represents the amount of the antibody produced on each day of sampling.

[0077] FIG. 3 The expression level of firefly luciferase transiently expressed using the region of approximately 2.5 kbp from approximately 5.0 kbp upstream from the start codon of a Chinese hamster-derived Eno1 gene as a promoter was compared between the chain lengths. The vertical axis represents a value obtained by dividing the amount of luminescence of firefly-derived luciferase on pGL 4.10 by the amount of luminescence of *Renilla*-derived luciferase on pGL 4.74. In the figure, Eno1 5k, Eno1 4.5k, Eno1 4k, Eno1 3.5k, Eno1 3.0k, Eno1 2.8k, Eno1 2.7k, Eno1 2.6k and Eno1

2.5k denote, respectively, regions extending to approximately 5.0 kbp (SEQ ID NO: 10), approximately 4.5 kbp (SEQ ID NO: 11), approximately 4.0 kbp (SEQ ID NO: 4), approximately 3.5 kbp (SEQ ID NO: 12), approximately 3.0 kbp (SEQ ID NO: 1), approximately 2.8 kbp (SEQ ID NO: 3), approximately 2.7 kbp (SEQ ID NO: 13), approximately 2.6 kbp (SEQ ID NO: 14) and approximately 2.5 kbp (SEQ ID NO: 15) upstream from a nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in the Chinese hamster-derived Eno1 gene.

[0078] FIG. 4-A The amount of antibody produced in the fed-batch culture of a humanized antibody Y-expressing stable pool generated using an Eno1 gene promoter derived from several species was compared. Eno1 and mEno1 indicate results from the Chinese hamster-derived Eno1 gene promoter and the mouse-derived Eno1 gene promoter, respectively. In FIG. 4-A, the vertical axis represents the number of viable cells on each day of sampling. In the figure, Eno1 4k, Eno1 3.5k, Eno1 3k and Eno1 2.8k denote, respectively, regions extending to approximately 4.0 kbp (SEQ ID NO: 4), approximately 3.5 kbp (SEQ ID NO: 12), approximately 3.0 kbp (SEQ ID NO: 1) and approximately 2.8 kbp (SEQ ID NO: 3) upstream from a nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in the Chinese hamster-derived Eno1 gene. mEno1 4k, mEno1 3.5k and mEno1 3k denote, respectively, regions extending to approximately 4.0 kbp (SEQ ID NO: 16), approximately 3.5 kbp (SEQ ID NO: 17) and approximately 3.0 kbp (SEQ ID NO: 2) upstream from a nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in the mouse-derived Eno1 gene (the same applies to FIG. 4-B).

[0079] FIG. 4-B The amount of antibody produced in the fed-batch culture of a humanized antibody Y-expressing stable pool generated using an Eno1 gene promoter derived from several species was compared. Eno1 and mEno1 indicate results from the Chinese hamster-derived Eno1 gene promoter and the mouse-derived Eno1 gene promoter, respectively. In FIG. 4-B, the vertical axis represents the amount of the antibody produced on each day of sampling.

[0080] FIG. 5-A The amount of antibody produced in the fed-batch culture of a humanized antibody Y-expressing stable pool generated using an Eno1 gene promoter derived from several species and DNA element A7 in combination was compared. Eno1 and mEno1 indicate results from the Chinese hamster-derived Eno1 gene promoter and the mouse-derived Eno1 gene promoter, respectively. In FIG. 5-A, the vertical axis represents the number of viable cells on each day of sampling. In the figure, the Eno1 gene promoters are set similarly to those in FIG. 4-A. A7(+) indicates combination with DNA element A7, and A7(-) indicates absence of A7 (the same applies to FIG. 5-B).

[0081] FIG. 5-B The amount of antibody produced in the fed-batch culture of a humanized antibody Y-expressing stable pool generated using an Eno1 gene promoter derived from several species and DNA element A7 in combination was compared. In FIG. 5-B, the vertical axis represents the amount of the antibody produced on each day of sampling.

[0082] FIG. 6 shows a sequence listing.

DESCRIPTION OF EMBODIMENTS

[0083] Hereinafter, preferred forms for carrying out the present invention will be described. The embodiments

described below show examples of typical embodiments of the present invention, by which the scope of the present invention should not be narrowly interpreted.

[0084] In the present description, the term “gene” means a moiety that is transcribed into mRNA and translated into a protein, and is used to include, not only DNA but also its mRNA and cDNA, and the RNA thereof.

[0085] In the present description, the term “polynucleotide” is used to have the same meaning as that of a nucleic acid, and includes DNA, RNA, a probe, an oligonucleotide, and a primer.

[0086] In the present description, the term “polypeptide” is used without being distinguished from the term “protein”.

[0087] In the present description, the term “gene expression” means a phenomenon in which a gene is transcribed into mRNA, and/or a phenomenon in which the mRNA is translated into a protein.

[0088] In the present description, the term “foreign gene” means a gene that is artificially introduced into host cells.

[0089] In the present description, the term “foreign protein” means a protein encoded by the foreign gene.

[0090] In the present description, the term “gene expression unit” means a polynucleotide having at least a promoter region, a foreign gene, and a transcriptional terminator region (polyA addition signal) in the reading frame direction of transcription.

[0091] In the present description, the term “promoter” means a region to which a transcription factor involved in the start of transcription of DNA into RNA binds. In the present description, the term “promoter region” is also used. Examples of the promoter can include a polynucleotide from a nucleotide approximately 3 kbp upstream of a start codon to a nucleotide immediately upstream of a nucleotide sequence corresponding to the start codon. The promoter may contain 5'UTR and an intron.

[0092] In the present description, the term “promoter activity” refers to activity by which transcription starts through the binding of a transcription factor to the promoter to perform the production of a protein encoded by the gene. Promoter activity can be examined by using, as an indicator, the activity of a protein encoded by a reporter gene, such as firefly luciferase.

[0093] In the present description, the term “having promoter activity” means the confirmation of antibody expression under conditions similar to those used to evaluate promoter activity indicated by the amount of antibody expressed in the fed-batch culture described in (Example 2), or the confirmation of activity of firefly luciferase under conditions similar to those described in (Example 3).

[0094] In the present description, the term “DNA element” means a polynucleotide having foreign gene expression-enhancing activity when located in proximity to a gene expression unit or located in a foreign gene expression vector comprising the gene expression unit.

[0095] In the present description, the term “antigen-binding fragment of the antibody” means a partial fragment of the antibody having binding activity to the antigen. Examples thereof include Fab and F(ab')₂, though the antigen-binding fragment is not limited to these molecules as long as it has antigen-binding ability.

[0096] In the present description, the term “sequence identity” refers to the relationship between sequences as to two or more nucleotide sequences or amino acid sequences and is determined by comparison of the sequences, as known

in the art. The term “sequence identity” in the art means, in some cases, the degree to which the sequence of a nucleic acid molecule or a polypeptide matches when comparing between two or more nucleotide sequences or between two or more amino acid sequences. The term “sequence identity” can be evaluated by calculating percentage identity between a smaller sequence of two or more sequences and gap alignment (if present) addressed by a particular mathematical model or computer program (i.e., “algorithm”). Specifically, the identity can be evaluated using software such as ClustalW2 provided by European Molecular Biology Laboratory-European Bioinformatics Institute (EMBL-EBI), though the evaluation method is not limited thereto as long as the method is used by a person skilled in the art.

[0097] In the present description, the phrase “to hybridize under stringent conditions” refers to hybridization under conditions in which a so-called specific hybrid is formed whereas a non-specific hybrid is not formed. Examples thereof can include conditions under which a complementary strand of a nucleic acid consisting of a nucleotide sequence having 80% or higher, preferably 90% or higher, more preferably 95% or higher, most preferably 99% or higher sequence identity to a nucleic acid hybridizes whereas a complementary strand of a nucleic acid consisting of a nucleotide sequence having lower sequence identity does not hybridize. More specifically, the phrase is used to mean that hybridization is carried out in the commercially available hybridization solution ExpressHyb Hybridization Solution (manufactured by Clontech) at 68° C., or that hybridization is carried out under conditions using a DNA-immobilized filter in the presence of 0.7 to 1.0 M NaCl at 68° C., and the resultant is then washed at 68° C. with a 0.1× to 2×SSC solution (wherein 1×SSC consists of 150 mM NaCl and 15 mM sodium citrate), or conditions equivalent thereto.

[0098] In the present description, the term “approximately” refers to a value that varies by up to 10%, 8%, 6%, 5%, 4%, 3%, 2% or 1% in either a plus or minus direction with respect to a reference value. Preferably, the term “approximately” refers to a range of 10%, 5% or 1% in either a plus or minus direction with respect to a reference value.

1. Promoter for Use in Enhancement of Foreign Gene Expression

[0099] The promoter for use in the enhancement of foreign gene expression according to the present invention (hereinafter, also referred to as the “promoter of the present invention”) is a promoter of an α -enolase (alpha-enolase) (hereinafter, referred to “Eno1”) gene.

[0100] The origin of the Eno1 gene promoter is not particularly limited and may be of mammalian origin. Examples thereof can include Chinese hamster, human, and mouse derived Eno1 gene promoters. The promoter of the present invention is preferably a rodent-derived Eno1 gene promoter, more preferably a Chinese hamster-, or mouse-derived Eno1 gene promoter. The promoter of the present invention is still more preferably a Chinese hamster-derived Eno1 gene promoter.

[0101] Examples of the Chinese hamster-derived Eno1 gene promoter include a sequence from a nucleotide that is, for example, 2.0 kbp, 2.5 kbp, 2.6 kbp, 2.7 kbp, 2.8 kbp, 3.0 kbp, 3.5 kbp, 4.0 kbp, 4.5 kbp, 5.0 kbp, 5.5 kbp or 6.0 kbp

upstream of the start codon of the Eno1 gene, to a nucleotide immediately upstream of a nucleotide sequence corresponding to the start codon.

[0102] More specifically, examples of the Chinese hamster-derived Eno1 gene promoter include the polynucleotides of SEQ ID NOs: 1, 3, 4, 10, 11, 12, 13 and 14.

[0103] The nucleotide sequence of SEQ ID NO: 1 is a sequence from a nucleotide approximately 3.0 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0104] The nucleotide sequence of SEQ ID NO: 3 is a sequence from a nucleotide approximately 2.8 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0105] The nucleotide sequence of SEQ ID NO: 4 is a sequence from a nucleotide approximately 4.0 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0106] The nucleotide sequence of SEQ ID NO: 10 is a sequence from a nucleotide approximately 5.0 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0107] The nucleotide sequence of SEQ ID NO: 11 is a sequence from a nucleotide approximately 4.5 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0108] The nucleotide sequence of SEQ ID NO: 12 is a sequence from a nucleotide approximately 3.5 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0109] The nucleotide sequence of SEQ ID NO: 13 is a sequence from a nucleotide approximately 2.7 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0110] The nucleotide sequence of SEQ ID NO: 14 is a sequence from a nucleotide approximately 2.6 kbp upstream of the start codon of the Chinese hamster-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0111] The Eno1 gene promoter can be a region having a length such that it extends to approximately 2.8 kbp or more upstream of the start codon sequence from a nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence because the promoter has excellent transcriptional activity. Therefore, from the viewpoint of excellent promoter activity, the Chinese hamster-derived Eno1 gene promoter is preferably the polynucleotide of SEQ ID NO: 1, 3, 4, 10, 11 or 12. The polynucleotide that can be selected as the Chinese hamster-derived Eno1 gene promoter is more preferably the polynucleotide of SEQ ID NO: 1, 3, 4 or 12, still more preferably the polynucleotide of SEQ ID NO: 1 or 3.

[0112] Examples of the mouse-derived Eno1 gene promoter include a sequence from a nucleotide that is, for example, 2.0 kbp, 2.5 kbp, 3.0 kbp, 3.5 kbp, 4.0 kbp, 4.5 kbp, 5.0 kbp, 5.5 kbp or 6.0 kbp upstream of the start codon

of the Eno1 gene, to a nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0113] More specifically, examples of the mouse-derived Eno1 gene promoter include the polynucleotides of SEQ ID NOS: 2, 17 and 16.

[0114] The nucleotide sequence of SEQ ID NO: 2 is a sequence from a nucleotide approximately 3.0 kbp upstream of the start codon of the mouse-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0115] The nucleotide sequence of SEQ ID NO: 17 is a sequence from a nucleotide approximately 3.5 kbp upstream of the start codon of the mouse-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0116] The nucleotide sequence of SEQ ID NO: 16 is a sequence from a nucleotide approximately 4.0 kbp upstream of the start codon of the mouse-derived Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon.

[0117] The promoter of the present invention may be a polynucleotide comprising a nucleotide sequence having 80% or higher, preferably 85% or higher, 86% or higher, 87% or higher, 88% or higher, 89% or higher, more preferably 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, still more preferably 95% or higher, 96% or higher, 97% or higher, 98% or higher, most preferably 99% or higher, or 100% sequence identity to the nucleotide sequence of any one of SEQ ID NOS: 1, 2, 17, 3, 4, 10, 11, 12, 13, 14 and 16, or a partial sequence of the nucleotide sequence, and having promoter activity.

[0118] The partial sequence may be a fragment whose length is 80% or more, preferably 85% or more, 86% or more, 87% or more, 88% or more, 89% or more, more preferably 90% or more, 91% or more, 92% or more, 93% or more, 94% or more, still more preferably 95% or more, 96% or more, 97% or more, 98% or more, most preferably 99% or more, or 100% of the full length of the nucleotide sequence of any one of SEQ ID NOS: 1, 2, 17, 3, 4, 10, 11, 12, 13, 14 and 16. The partial sequence may be a fragment on the 5' side (upstream) or the 3' side (downstream) of the full-length sequence, or may be a fragment obtained by connecting a plurality of fragments of the full-length sequence.

[0119] The promoter of the present invention may be a polynucleotide that hybridizes under stringent conditions to a polynucleotide consisting of a nucleotide sequence complementary to a polynucleotide consisting of the nucleotide sequence of any one of SEQ ID NOS: 1, 2, 17, 3, 4, 10, 11, 12, 13, 14 and 16, and has promoter activity.

[0120] The promoter of the present invention preferably comprises the nucleotide sequence of SEQ ID NO: 18, or a nucleotide sequence having 85% or higher sequence identity, more preferably 90% or higher, 91% or higher, 92% or higher, 93% or higher, or 94% or higher sequence identity, still more preferably 95% or higher, 96% or higher, 97% or higher, or 98% or higher sequence identity, most preferably 99% or higher, or 100% sequence identity to the nucleotide sequence.

[0121] In particular, the Chinese hamster-derived Eno1 promoter preferably comprises the nucleotide sequence of SEQ ID NO: 18, or a nucleotide sequence having 85% or higher sequence identity, more preferably 90% or higher, 91% or higher, 92% or higher, 93% or higher, or 94% or

higher sequence identity, still more preferably 95% or higher, 96% or higher, 97% or higher, or 98% or higher sequence identity, most preferably 99% or higher, or 100% sequence identity to the nucleotide sequence.

[0122] The length of the promoter of the present invention is not particularly limited as long as it is 1 kbp or more and 10 kbp or less, and the length of the promoter is preferably 2 kbp or more and 6 kbp or less, more preferably 2.6 kbp or more and 5 kbp or less, still more preferably 2.7 kbp or more and 4.5 kbp or less, particularly preferably 2.8 kbp or more and 4 kbp or less, and can be typically approximately 3 kbp. In the case of a promoter length ensuring sufficient promoter performance, it becomes easier to exhibit an effect in terms of transfection efficiency as the promoter length decreases.

[0123] The promoter of the present invention may be a modified polynucleotide consisting of a nucleotide sequence comprising a deletion, substitution, and/or addition of one or more, preferably 1 to 300, more preferably 1 to 200, still more preferably 1 to 100, particularly preferably 1 to 30 nucleotides in the nucleotide sequence of any one of SEQ ID NOS: 1, 2, 17, 3, 4, 10, 11, 12, 13, 14 and 16.

[0124] The introduction of a modification (deletion, substitution, and/or addition) into the nucleotide sequence can be performed by an approach known in the art such as the Kunkel method or the Gapped-Duplex method, or a method equivalent thereto. For example, a kit for mutation introduction which exploits site-directed mutagenesis (e.g., Mutant-K (manufactured by Takara Bio Inc.) or Mutant-G (manufactured by Takara Bio Inc.), or an LA PCR in vitro Mutagenesis series kit from Takara Bio Inc.) can be utilized. Such a modified polynucleotide can also be used as the promoter of the present invention.

[0125] The foreign gene expression-enhancing activity possessed by the promoter of the present invention can be examined by using, as an indicator, the activity of a protein encoded by a reporter gene, such as firefly luciferase, or the amount of an antibody produced in fed-batch culture.

[0126] When the activity of a protein encoded by a reporter gene of a vector comprising the promoter of the present invention (e.g., the amount of luminescence of firefly luciferase) is 1 or more times, preferably 20 or more times, more preferably 30 or more times, still more preferably 40 or more times, even more preferably 45 or more times, particularly preferably 50 or more times higher compared with the activity of a protein encoded by a reporter gene of a control vector (e.g., the amount of luminescence of *Renilla* luciferase), it can be determined that this promoter has foreign gene expression-enhancing activity.

[0127] When the amount of the antibody produced in fed-batch culture is equivalent or higher, preferably 1.2 or more times, more preferably 1.3 or more times, still more preferably 1.5 or more times, particularly preferably 1.8 or more times, 1.9 or more times, or 2.0 or more times by use of the promoter of the present invention compared with use of a human EF-1 α promoter, it can be determined that this promoter has foreign gene expression-enhancing activity. Even in the case of an enhancement of approximately 1.2-fold or more, a reduction of cell culture scale, culture time and the number of purification steps is expected. As a result, an improvement in yield and a reduction in culture cost are attained. The improved yield permits stable supply of a foreign protein as a medicament. Also, the reduced culture cost leads to a reduction in the prime cost of a foreign protein as a medicament.

2. Foreign Gene Expression Unit

[0128] The foreign gene expression unit of the present invention (hereinafter, also referred to as the “gene expression unit of the present invention”) has at least the promoter of the present invention described in the preceding section 1., a foreign gene, and a transcriptional terminator region (polyA addition signal) in the reading frame direction of transcription.

[0129] The polyA addition sequence can be any sequence having the activity of terminating transcription from the promoter, and may be derived from a gene which is the same as or different from the gene of the promoter.

3. DNA Element for Use in Enhancing Foreign Gene Expression

[0130] Combined use of the gene expression unit of the present invention described in the preceding section 2. with a DNA element can further enhance the expression of a foreign gene. The DNA element for combined use can be obtained by interaction with acetylated histone H3 as an indicator. In general, the acetylation of histone (H3, H4) is reportedly involved in the activation of transcription on the basis of two main hypotheses: that conformational change of the nucleosome is involved such that histone tail acetylation neutralizes the charge thereof to loosen the binding between the DNA and the histone (Mellor J. (2006), *Dynamic nucleosomes and gene transcription*, *Trends Genet.* 22 (6): 320-329); and that the acetylation is involved in the recruitment of various transcription factors (Nakatani Y. (2001), *Histone acetylases-versatile players*, *Genes Cells.* 6 (2): 79-86). Both of the hypotheses strongly suggest that the acetylation of histone is involved in transcriptional activation. Thus, chromatin immunoprecipitation (ChIP) using an anti-acetylated histone H3 antibody is capable of enriching a sample for a DNA element that interacts with acetylated histone H3.

[0131] Examples of the DNA element for use in the enhancing of foreign gene expression in combination with the promoter of the present invention can include A2, A7, and A18. The DNA element is preferably A7.

[0132] A2 is positioned at a site from 80966429 to 80974878 of human chromosome 15 and is an 8450 bp polynucleotide having an AT content of 62.2%. The nucleotide sequence of A2 is shown in SEQ ID NO: 5 in the sequence listing.

[0133] A7 is positioned at a site from 88992123 to 89000542 of human chromosome 11 and is an 8420 bp polynucleotide having an AT content of 64.52%. The nucleotide sequence of A7 is shown in SEQ ID NO: 6 in the sequence listing.

[0134] A18 is positioned at a site from 111275976 to 111284450 of human chromosome 4 and is an 8475 bp polynucleotide having an AT content of 62.54%. The nucleotide sequence of A18 is shown in SEQ ID NO: 7 in the sequence listing.

[0135] The foreign gene expression-enhancing activity possessed by the DNA element for combined use with the promoter of the present invention can be examined by using, as an indicator, the activity of a protein encoded by a reporter gene, such as the secreted alkaline phosphatase (SEAP).

[0136] For combined use with the promoter of the present invention, any one of the DNA elements described above may be used alone, or two or more copies of one DNA

element may be used. Alternatively, two or more DNA elements may be used in combination.

[0137] The DNA element used in the present invention may consist of a nucleotide sequence having 80% or higher, preferably 85% or higher, 86% or higher, 87% or higher, 88% or higher, or 89% or higher, more preferably 90% or higher, 91% or higher, 92% or higher, 93% or higher, or 94% or higher, still more preferably 95% or higher, 96% or higher, 97% or higher, 98% or higher, most preferably 99% or higher, or 100% identity to the nucleotide sequence of any one of SEQ ID NOs: 5 to 7, and having foreign gene expression-enhancing activity.

[0138] The DNA element used in the present invention may be a modified polynucleotide consisting of a nucleotide sequence comprising a deletion, substitution, and/or addition of one or more, preferably 1 to 1000 or 1 to 500, more preferably 1 to 100 or 1 to 50, still more preferably 1 to 10 or 1 to 5, particularly preferably 4, 3, 2 or 1 nucleotides in the nucleotide sequence of any one of SEQ ID NOs: 5 to 7, and having foreign gene expression-enhancing activity.

[0139] The introduction of a modification (deletion, substitution, and/or addition) into the polynucleotide can be performed by an approach known in the art such as the Kunkel method or the Gapped-Duplex method, or a method equivalent thereto. For example, a kit for mutation introduction (e.g., Mutant-K (manufactured by Takara Bio Inc.) or Mutant-G (manufactured by Takara Bio Inc.), or an LA PCR in vitro Mutagenesis series kit from Takara Bio Inc.) can be utilized which exploits site-directed mutagenesis. Such a mutant polynucleotide can also be used as the DNA element of the present invention.

[0140] As the DNA element used in the present invention, a partial sequence consisting of at least 3000 or at least 2000 consecutive nucleotides in the polynucleotide sequence shown in any one of SEQ ID NOs: 5 to 7 can be used. Such a partial sequence may be, for example, a partial sequence described in International Publication No. WO 2012/005378. Examples of the partial sequence A2 can include A2-1 to A2-17, examples of the partial sequence A7 can include A7-1 to A7-18, and examples of the partial sequence A18 can include A18-1 to A18-4.

[0141] In the present invention, any one of the partial sequences of the DNA element described above may be used alone, or two or more copies of one partial fragment may be used. Alternatively, two or more partial fragments may be used in combination. Full-length sequences and partial sequences of the DNA elements may be used in combination. In the combination, the partial element may be derived from a DNA element of a full-length sequence forming the combination, or may be derived from a DNA element of a different full-length sequence.

4. Obtaining the Polynucleotide

[0142] In the present invention, a polynucleotide comprising a foreign gene encoding a foreign protein whose production is to be enhanced as mentioned later can be obtained by a general method given below. The polynucleotide can be isolated, for example, by screening a cDNA library derived from cells or tissues expressing the foreign gene, using a DNA probe synthesized on the basis of a gene fragment. mRNA can be prepared by an approach usually used in the art. For example, the cells or the tissues are treated with a guanidine reagent, a phenol reagent, or the like to obtain total RNA. Then, poly(A)⁺ RNA (mRNA) is obtained there-

from by an affinity column method using an oligo (dT) cellulose column, polyU-Sepharose with Sepharose 2B as a carrier, or the like, or by a batch method. The poly(A)⁺ RNA may be further fractionated by a sucrose density gradient centrifugation method or the like. Subsequently, single-stranded cDNA is synthesized with the obtained mRNA as a template using an oligo dT primer and reverse transcriptase. Double-stranded cDNA is synthesized from the single-stranded cDNA using DNA synthetase I, DNA ligase and RNase H, etc. The synthesized double-stranded cDNA is converted into a blunt-ended form with T4 DNA synthetase, then subjected to the linkage of an adaptor (e.g., an EcoRI adaptor), phosphorylation, etc., and incorporated into a λ phage such as λ gt11 for in vivo packaging to prepare a cDNA library. Alternatively, the cDNA library may be prepared using a plasmid vector instead of the λ phage. Then, a clone having the DNA of interest (a positive clone) can be selected from the cDNA library.

[0143] In the case of isolating a polynucleotide comprising the promoter and a terminator region, the DNA element, or a polynucleotide comprising a foreign gene for use in protein production from genomic DNA, the genomic DNA is extracted from a cell line of an organism serving as a source, followed by polynucleotide selection, according to a general approach (Molecular Cloning (1989) and Methods in Enzymology 194 (1991)). The extraction of the genomic DNA can be performed according to, for example, the method of Cryer et al. (Methods in Cell Biology, 12, 39-44 (1975)) and the method of P. Philippsen et al. (Methods Enzymol., 194, 169-182 (1991)).

[0144] The obtaining of the polynucleotide of interest comprising the promoter, the DNA element, or a foreign gene can also be performed by, for example, PCR (PCR Technology, Henry A. Erlich, Atackton press (1989)). The amplification of the polynucleotide by PCR employs a 20 to 30 mer synthetic single-stranded DNA as a primer and genomic DNA as a template. The amplified gene is used after its polynucleotide sequence is confirmed. A genomic DNA library such as a bacterial artificial chromosome (BAC) library may be used as a template for PCR.

[0145] On the other hand, a polynucleotide comprising a foreign gene having an unknown sequence can be obtained by (a) preparing a gene library according to a common method, (b) selecting the desired polynucleotide from the prepared gene library, and amplifying the polynucleotide. The gene library can be prepared by partially digesting chromosomal DNA obtained by a common method from a cell line of an organism serving as a source, with an appropriate restriction enzyme to prepare fragments, ligating the obtained fragments into an appropriate vector, and introducing the vector into an appropriate host. Alternatively, the gene library may be prepared by extracting mRNA from the cells, synthesizing cDNA therefrom, then ligating the cDNA into an appropriate vector, and introducing the vector into an appropriate host. In this respect, a plasmid known as a well-known vector for gene library preparation can be used as the vector, and a phage vector or a cosmid, etc. can also be widely used. The host to be transfected or transduced can be used according to the type of vector. The polynucleotide comprising a foreign gene is selected from the gene library by colony hybridization, plaque hybridization, or the like using a labeled probe comprising a sequence unique to the foreign gene.

[0146] The polynucleotide comprising a foreign gene can also be synthesized entirely chemically. The gene can be synthesized by, for example, a method of preparing two pairs of complementary oligonucleotides and annealing these, a method of ligating several annealed DNAs using DNA ligase, or a method of preparing several partially complementary oligonucleotides and filling the gaps therein by PCR.

[0147] The polynucleotide sequence can be determined by a usual method, for example, the dideoxy method (Sanger et al., Proc. Natl. Acad. Sci., USA, 74, 5463-5467 (1977)). Alternatively, the polynucleotide sequence may be readily determined using a commercially available sequencing kit or the like.

5. Foreign Gene Expression Vector

[0148] A vector comprising the foreign gene expression unit described in the preceding section 2. comprising the promoter described in the preceding section 1. is provided as a foreign gene expression vector of the present invention. The foreign gene expression vector of the present invention may comprise one of the DNA elements described in the preceding section 3., two or more copies of one DNA element, or a combination of two or more DNA elements. When a foreign gene is expressed in host cells using the foreign gene expression vector, the DNA element may be located immediately preceding or immediately following the gene expression unit, or may be located at a position distant from the gene expression unit. Alternatively, one foreign gene expression vector comprising a plurality of DNA elements may be used. The orientation of the DNA element may be in either the forward direction or the reverse direction with respect to the gene expression unit.

[0149] Examples of the foreign gene can include, but are not particularly limited to: reporter genes such as the secreted alkaline phosphatase (SEAP) gene, the green fluorescence protein (GFP) gene, and the luciferase gene; various enzyme genes such as the α -amylase gene and the α -galactosidase gene; genes of various interferons such as interferon α and interferon γ , which are pharmaceutically useful physiologically active proteins; genes of various interleukins such as IL1 and IL2; various cytokine genes such as the erythropoietin (EPO) gene and the granulocyte colony-stimulating factor (G-CSF) gene; growth factor genes; and a gene encoding a multimeric protein, for example, a gene encoding a heteromultimer which is an antibody or an antigen-binding fragment thereof. These genes may be obtained by any approach.

[0150] The term "antigen-binding fragment of the antibody" means a partial fragment of the antibody having binding activity to the antigen. Examples thereof include Fab, F(ab')₂, Fv, scFv, a diabody, linear antibodies, and multispecific antibodies formed from antibody fragments. Also, Fab', which is a monovalent fragment of antibody variable regions obtained by the treatment of F(ab')₂ under reductive conditions is included within the antigen-binding fragment of the antibody. However, the antigen-binding fragment is not limited to these molecules as long as it has antigen-binding ability. Furthermore, these antigen-binding fragments also include not only a fragment obtained by the treatment of the full-length molecule of an antibody protein with an appropriate enzyme but also a protein produced in appropriate host cells using a genetically engineered antibody gene.

[0151] The foreign gene expression vector of the present invention can comprise a selection marker for selecting a transfectant. The transfectant can be selected using, for example, a drug resistance marker which confers resistance to a drug such as cerulenin, aureobasidin, zeocin, canavanine, cycloheximide, hygromycin, puromycin, blasticidin, tetracycline, kanamycin, ampicillin, or neomycin. Alternatively, the transfectant may be selected by using, as a marker, a gene that confers, for example, solvent resistance to ethanol or the like, osmotic pressure resistance to glycerol, a salt, or the like, or metal ion resistance to copper or the like.

[0152] The foreign gene expression vector of the present invention may be a vector that is not integrated into chromosomal DNA. In general, the foreign gene expression vector is randomly integrated into the chromosome after transfection of host cells. By contrast, use of a constituent derived from a mammalian virus such as simian virus 40 (SV40), papillomavirus (BPV, HPV), or EBV allows the foreign gene expression vector to be used as an episomal vector capable of replicating autonomously in the transfected host cells. For example, a vector having a sequence encoding a SV40-derived replication origin and a transacting factor SV40 large T antigen, or a vector having a sequence encoding EBV-derived oriP and EBNA-1 is widely used. The DNA element is capable of exhibiting foreign gene expression-enhancing activity, regardless of the type of vector or the presence or absence of integration into the chromosome.

6. Transfected Cells

[0153] The transfected cells of the present invention are transfected cells comprising the foreign gene expression vector of the preceding section 5. introduced thereinto.

[0154] The host cells to be transfected are eukaryotic cells, preferably mammalian cells, more preferably human-, mouse-, rat-, hamster-, monkey-, or bovine-derived cells. Examples of the mammalian cells can include, but are not limited to, COS-1 cells, 293 cells, and CHO cells (e.g., CHO-K1, CHO-01, CHO DG44, CHO dhfr-, CHO—S).

[0155] In the present invention, the method for introducing the expression vector into host cells can be any method as long as the method allows the introduced gene to be present stably in the host and to be appropriately expressed. Examples thereof can include methods generally used, for example, the calcium phosphate method (Ito et al., (1984) *Agric. Biol. Chem.*, 48, 341), electroporation (Becker, D. M. et al. (1990), *Methods. Enzymol.*, 194, 182-187), the spheroplast method (Creggh et al., *Mol. Cell. Biol.*, 5, 3376 (1985)), the lithium acetate method (Itoh, H. (1983), *J. Bacteriol.* 153, 163-168), and lipofection.

[0156] In the transfected cells according to the present invention, a foreign gene may be transiently expressed, or stably expressed, but is preferably expressed in a stable expression system.

[0157] The term “stable expression system” refers to a method in which an expression vector is incorporated into a chromosome by the calcium phosphate method, electroporation, lipofection or the like to perform the expression. The introduced gene is maintained on the chromosome, so that the expression of the introduced gene can be maintained for a long period of several weeks or longer. Introduction of a selection marker into plasmids enables drug selection. It is

possible to efficiently select cells in which the introduced gene is maintained on the chromosome.

7. Method for Producing Foreign Protein

[0158] The production of a foreign protein according to the present invention can be performed by culturing the transfected cells described in the preceding section 6. by a known method, and collecting the foreign protein from the culture, followed by purification. The “culture” means any of a culture supernatant, cultured cells, and a cell homogenate. Not only a monomeric protein but also a multimeric protein may be selected as the foreign protein that can be produced using the transfected cells described in section 6. In the case of producing a heteromultimeric protein constituted by a plurality of different subunits, a plurality of genes encoding these subunits each need to be introduced into the host cells described in section 6.

[0159] The transfected cells can be cultured according to usual methods for use in the culture of the host cells.

[0160] When the transfected cells are mammalian cells, the transfected cells are cultured, for example, at 37° C. under 5% or 8% CO₂ conditions for a culture time in the order of 24 to 1000 hours. The culture can be carried out by, for example, static culture, shake culture, stirring culture, batch culture under aeration, fed-batch culture, perfusion culture or continuous culture. The culture is preferably fed-batch culture. The period of the fed-batch culture of the transfected cells may be 3 days or more and 20 days or less, and is preferably 5 days or more and 18 days or less, particularly preferably 6 days or more and 15 days or less.

[0161] The expression product of the foreign protein gene from the culture (culture solution) described above can be confirmed by SDS-PAGE, Western blotting, ELISA, or the like.

8. Method for Producing Antibody Protein

[0162] Examples of the heteromultimeric protein to be produced using the production method described in the preceding section 7. can include antibody proteins. The antibody protein is a tetramer protein consisting of two molecules of a heavy chain polypeptide and two molecules of a light chain polypeptide. Thus, for obtaining an antibody protein in a form that maintains antigen-binding ability, it is necessary to introduce both, heavy chain and light chain genes, into the transfected cells described in the preceding section 6. In this case, heavy chain and light chain gene expression units may be present on the same expression vector or may be present on different expression vectors. When the gene expression units are present on the same expression vector, the light chain gene expression unit and the heavy chain gene expression unit may be present in this order, and it is preferable that the DNA element, the light chain gene expression unit and the heavy chain gene expression unit be present in this order, in a gene reading direction. More preferably, the DNA element, the light chain gene expression unit and the heavy chain gene expression unit present in this order are followed by a selection marker.

[0163] Examples of the antibody to be produced according to the present invention can include antibodies prepared by immunizing laboratory animals such as rabbits, mice, and rats with the desired antigen. Further examples of the antibody to be produced according to the present invention can include chimeric antibodies and humanized antibodies

originating from the antibodies described above. In addition, a human antibody obtained from a genetically engineered animal or by the phage display method is also an antibody to be produced according to the present invention.

[0164] The antibody gene for use in antibody production is not limited to an antibody gene having a particular polynucleotide sequence as long as a combination of a heavy chain polypeptide and a light chain polypeptide obtained by the transcription of the antibody genes and subsequent translation retains the activity of binding to a given antigen protein.

[0165] The antibody gene is not necessarily required to encode the full-length molecule of the antibody. A gene encoding an antigen-binding fragment of the antibody can be used. The genes encoding such an antigen-binding fragment can be obtained by genetically engineering genes encoding the full-length molecule of the antibody protein.

9. Methods for Producing Other Foreign Proteins

[0166] Examples of the foreign protein to be produced by the production method of the present invention can include the antibodies mentioned above as well as various human- or non-human animal-derived proteins, antigen-binding fragments thereof, and modified forms of the proteins or the fragments. Examples of such a protein and the like can include, but are not limited to: peptide hormones such as atrial natriuretic peptide (ANP), brain natriuretic peptide (BNP), C-type natriuretic peptide (CNP), vasopressin, somatostatin, growth hormone (GH), insulin, oxytocin, ghrelin, leptin, adiponectin, renin, calcitonin, osteoprotegerin, and insulin-like growth factor (IGF); cytokines such as interleukin, chemokine, interferon, tumor necrosis factor (TNF α / β as well as TNF superfamily, etc.), nerve growth factor (NGF), cell growth factor (EGF, FGF, PDGF, HGF, TGF, etc.), hematopoietic factor (CSF, G-CSF, erythropoietin, etc.), and adiponectin; receptors such as TNF receptor; enzymes such as lysozyme, protease, proteinase, and peptidase; functional fragments thereof (fragments partially or wholly retaining the biological activity of the original protein); and fusion proteins comprising these proteins.

EXAMPLES

[0167] Hereinafter, the present invention will be specifically described with reference to Examples. However, these Examples do not limit the technical scope of the present invention by any means. Plasmids, restriction enzymes, DNA-modifying enzymes, etc. used in the Examples of the present invention are commercially available and can be used according to common methods. Operations used in DNA cloning, polynucleotide sequencing, transfection of host cells, culture of transfected cells, collection of a protein from the resulting culture, purification, etc. are also well known to a person skilled in the art or can be derived from the literature.

(Example 1) Cloning of an Eno1 Gene Promoter Region

[0168] The Eno1 promoter region used was a sequence from a nucleotide approximately 3.0 kbp upstream of the start codon sequence of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence with reference to the sequence of the mRNA registered under XM_027398299.1

and the scaffold sequence of the Chinese hamster genome registered under NC_048595.1 in GenBank.

[0169] The Eno1 gene promoter region was amplified by PCR with the genomic DNA of CHO cells as a template using the primer set given below and PrimeSTAR Max DNA Polymerase (Takara Bio Inc.), and the PCR product was purified using a QIAquick PCR Purification kit (Qiagen). The nucleotide sequence of the cloned Chinese hamster Eno1 gene promoter region is shown in SEQ ID NO: 1 in the sequence listing.

Primer Set for the Eno1 Gene Promoter

K24 (Fw) : (SEQ ID No. 8)
 TTCGCGCGCCGCGGCACGCAGCGCGCAGG
 K25 (Rev) : (SEQ ID NO. 9)
 TTCCTAGTTGTCTGTAGGGGAAAAAAAAC

(Example 2) Evaluation of the Chinese Hamster-Derived Eno1 Gene Promoter by Fed-Batch Culture with Antibody Expression Levels as Indicator

(2-1) Construction of Antibody Expression Vector

[0170] As an antibody expression vector, pDSLH3.1-Eno1-Y was constructed by substituting a nucleotide sequence corresponding to the Hspa5 gene promoter of humanized antibody gene Y expression vector pDSLH3.1-Hspa5-Y described in Patent Literature 4 with the nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter using the DNA fragment amplified and purified in Example 1. The vectors are schematically shown in FIG. 1. pDSLH3.1-hEF1 α -Y used for comparison is as described in Patent Literature 4.

(2-2) Generation of a Humanized Antibody Y-Expressing Stable Pool

[0171] CHO-K1 cells (ATCC) were adapted to suspension culture in a serum-free medium, to obtain CHO-01 cells as host cells. The CHO-01 cells were transfected with the antibody expression vector pDSLH3.1-Eno1-Y constructed in (2-1) or pDSLH3.1-hEF1 α -Y described in Patent Literature 4 using a transfection apparatus Neon Transfection System (Invitrogen), and cultured in 5% CO₂ at 37° C. in a T-25 flask. One day after the transfection, Geneticin (Life Technologies Corporation) was added thereto at a final concentration of 800 μ g/mL, followed by a drug selection culture for 1 week. Then, the cells were cultured in 5% CO₂ at 37° C. in a 125 mL Erlenmeyer flask to obtain a humanized antibody Y-expressing stable pool (two pools for each vector).

(2-3) Evaluation of the Amount of Antibody Produced by Fed-Batch Culture of Humanized Antibody Y-Expressing Stable Pools

[0172] Fed-batch culture was performed for 14 days in a 125 mL Erlenmeyer flask using each humanized antibody Y-expressing stable pool generated in (2-2). The culture conditions were as follows: the initial seeding density is 0.3×10^6 cells/mL, the amount of the culture solution is 30

mL, the basal medium used is G13 (custom medium manufactured by FUJIFILM Wako Pure Chemical Corporation), and the feed medium used is F13 (custom medium manufactured by FUJIFILM Wako Pure Chemical Corporation). The feed medium was added in an amount of 3% (v/v) of the initial culture solution amount every day from day 3 to day 13 of culture. On days 7, 10 and 14, sampling was performed, and the viable cell density, the amount of antibody produced, and the like were measured.

[0173] The change in the number of viable cells and the change in the amount of antibody produced are shown in FIGS. 2A and 2B, respectively. An average value of the amounts of antibodies produced in two pools generated for each vector, and the antibody productivity was compared. The results showed that the amount of antibody produced with the Eno1 promoter on day 14 of the culture reached 2.0 times the value of that with the human EF1- α promoter, and thus greatly exceeded the amount of antibody produced with a promoter that is currently frequently used.

(Example 3) Study on Chinese Hamster-Derived Eno1 Gene Promoter Length with the Amount of Luminescence of Luciferase as an Indicator in Transient Expression

(3-1) Construction of Luciferase Expression Vectors

[0174] Eno1 promoters with respective chain lengths were inserted into pGL4.10 (manufactured by Promega Corporation) to construct firefly-derived luciferase expression vectors. The chains whose length was studied were those extending to approximately 5.0 kbp (SEQ ID NO: 10), approximately 4.5 kbp (SEQ ID NO: 11), approximately 4.0 kbp (SEQ ID NO: 4), approximately 3.5 kbp (SEQ ID NO: 12), approximately 3.0 kbp (SEQ ID NO: 1), approximately 2.8 kbp (SEQ ID NO: 3), approximately 2.7 kbp (SEQ ID NO: 13), approximately 2.6 kbp (SEQ ID NO: 14) and approximately 2.5 kbp (SEQ ID NO: 15) upstream from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in the Chinese hamster-derived Eno1 gene.

(3-2) Evaluation of the Amount of Luminescence of Luciferase in Transient Expression

[0175] Using Lipofectamin 2000 CD (manufactured by Thermo Fisher Scientific), CHO-01 cells were transfected simultaneously with the firefly-derived luciferase expression vector constructed in (3-1) and a control vector pGL4.74 that expresses *Renilla*-derived luciferase. Twenty four hours after the transfection, the cells were centrifuged (1000 rpm, 3 min), collected in a 1.5 mL tube, and washed with PBS buffer. Then, the amount of luminescence of luciferase was measured using a Dual-Luciferase Reporter Assay System (manufactured by Promega Corporation). For each promoter length, transfection was performed with N=2.

[0176] The value obtained by dividing the amount of luminescence of firefly-derived luciferase on pGL 4.10 by the amount of luminescence of *Renilla*-derived luciferase on pGL 4.74 was taken as promoter activity (FIG. 3). Table 1 shows average values calculated from promoter activity values obtained with N=2. The results show that the promoter activity obtained using any of the regions extending to approximately 5.0 kbp (SEQ ID NO: 10), approximately 4.5 kbp (SEQ ID NO: 11), approximately 4.0 kbp (SEQ ID NO: 4) and approximately 3.5 kbp (SEQ ID NO: 12) upstream of

the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in the Chinese hamster-derived Eno1 gene was equal to or higher than 80% of the promoter activity obtained using a region extending to approximately 3.0 kbp (SEQ ID NO: 1) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in the Chinese hamster-derived Eno1 gene. The promoter activity obtained using a region extending to 2.8 kbp (SEQ ID NO: 3) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence was substantially equal to the promoter activity obtained using a region extending to approximately 3.0 kbp (SEQ ID NO: 1) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence. The promoter activity obtained using any of the regions extending to 2.7 kbp (SEQ ID NO: 13) and 2.6 kbp (SEQ ID NO: 14) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence was approximately 50% of the promoter activity obtained using a region extending to approximately 3.0 kbp (SEQ ID NO: 1) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence. The promoter activity obtained using a region extending to 2.5 kbp (SEQ ID NO: 15) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence was approximately 15% of the promoter activity obtained using a region extending to approximately 3.0 kbp (SEQ ID NO: 1) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence. Thus, it was found that the transcriptional activity of the Chinese hamster-derived Eno1 gene promoter was markedly high in a region having a length such that it extends to approximately 2.8 kbp or more upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence.

[0177] In the region between SEQ ID NO: 3 (Eno1 2.8 k) and SEQ ID NO: 13 (Eno1 2.7 k), the transcriptional activity of the promoter was markedly higher in SEQ ID NO: 3 (Eno1 2.8 k). For this reason, it was considered that when the promoter had a differential sequence between SEQ ID NO: 18 (differential sequence between SEQ ID NO: 3 (Eno1 2.8 k) and SEQ ID NO: 13 (Eno1 2.7 k) (100 bp)), its transcriptional activity was particularly improved.

TABLE 1

Promoter	Promoter Activity (Firefly Luc/ <i>Renilla</i> Luc)
Eno1 5 k	1.1×10^2
Eno1 4.5 k	1.0×10^2
Eno1 4 k	93
Eno1 3.5 k	88
Eno1 3.0 k	1.1×10^2
Eno1 2.8 k	1.0×10^2
Eno1 2.7 k	56
Eno1 2.6 k	48
Eno1 2.5 k	16

(Example 4) Evaluation of the Mouse-Derived
Eno1 Gene Promoter by Fed-Batch Culture with
Antibody Expression Levels as Indicator

(4-1) Construction of Antibody Expression Vector

[0178] pDSLH3.1-mEno1-Y, pDSLH3.1-mEno1-3.5-Y, and pDSLH3.1-mEno1-4.0-Y were constructed by substituting the promoter for antibody H chain and L chain genes in the humanized antibody gene Y expression vector pDSLH3.1-hEF1 α -Y with the mouse Eno1 promoter. For the mouse-derived Eno1 gene promoter, a region extending to approximately 3.0 kbp (SEQ ID NO: 2), approximately 3.5 kbp (SEQ ID NO: 17) or approximately 4.0 kbp (SEQ ID NO: 16) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in Eno1.

[0179] Similarly, regions extending to approximately 4.0 kbp (SEQ ID NO: 4), approximately 3.5 kbp (SEQ ID NO: 12) and approximately 2.8 kbp (SEQ ID NO: 3) upstream of the start codon sequence from the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon sequence in the Chinese hamster-derived Eno1 gene promoter were used as the Eno1 gene promoter to construct pDSLH3.1-Eno1-4.0-Y, pDSLH3.1-Eno1-3.5-Y and pDSLH3.1-Eno1-2.8-Y, respectively.

(4-2) Generation of Humanized Antibody Y-Expressing
Stable Pools

[0180] The CHO-01 cells were transfected with pDSLH3.1-hEF1 α -Y (Patent Literature 4), DSLH3.1-Eno1-Y constructed in (2-1), and pDSLH3.1-mEno1-Y, pDSLH3.1-mEno1-3.5-Y, pDSLH3.1-mEno1-4.0-Y, pDSLH3.1-Eno1-2.8-Y, pDSLH3.1-Eno1-3.5-Y and pDSLH3.1-Eno1-4.0-Y constructed in (4-1) by the method described in 2-2). Drug selection culture was performed to generate a humanized antibody Y-expressing stable pool (two pools for each vector).

(4-3) Evaluation of the Amount of Antibody Produced by
Fed-Batch Culture of Humanized Antibody Y-Expressing
Stable Pools

[0181] Fed-batch culture was performed in a 125 mL Erlenmeyer flask using each humanized antibody Y-expressing stable pool generated in (4-2). The basal medium used was G13, and the feed medium used was F13.

[0182] The change in the number of viable cells and the change in the amount of antibody produced are shown in FIGS. 4-A and 4-B, respectively. An average value of the amounts of antibody produced in two pools generated for each vector, and the antibody productivities were compared. The results showed that the amounts of antibody produced with the mouse-derived Eno1 gene promoters 3.0 kbp (mEno1 3k), 3.5 kbp (mEno1 3.5k) and 4.0 kbp (mEno1 4k) on day 14 of culture reached, respectively, 1.3 times, 1.8 times and 1.9 times the value of that with the human EF1 α promoter (hEF1 α), and thus greatly exceeded the amount of antibody produced with a promoter that is currently frequently used. It was found that the mouse-derived Eno1 gene promoter was equivalent in promoter activity to the Chinese hamster-derived Eno1 gene promoter.

[0183] The amounts of antibody produced with the Chinese hamster-derived Eno1 gene promoters 2.8 kbp (Eno1 2.8k), 3.0 kbp (Eno1 3k), 3.5 kbp (Eno1 3.5k) and 4.0 kbp

(Eno1 4k) reached, respectively, 2.7 times, 2.0 times, 2.8 times and 3.1 times the value of that with the human EF1 α promoter (on day 14 of culture), and thus greatly exceeded the amount of antibody produced with a promoter that is currently frequently used.

[0184] It was found that the Chinese hamster-derived Eno1 gene promoter was superior in antibody productivity to the mouse-derived Eno1 gene promoter.

[0185] The amount of antibody produced per cell per day (Qp) was calculated from the following equation.

[0186] $Qp = [\text{amount of antibody produced on day 10}] + [\text{integral viable cell density on day 10}]$ (the “integral viable cell density on day 10” is an integral value of time-dependent changes in viable cell density, which were calculated using the viable cell density on days 0, 7 and 10).

[0187] The amounts of antibody produced with the Chinese hamster-derived Eno1 gene promoters 2.8 kbp (Eno1 2.8k), 3.0 kbp (Eno1 3k), 3.5 kbp (Eno1 3.5k) and 4.0 kbp (Eno1 4k) reached, respectively, 2.7 times, 2.3 times, 3.0 times and 3.1 times the value of that with the human EF1 α promoter.

[0188] The amounts of antibody produced with the mouse-derived Eno1 gene promoters 3 kbp (mEno1 3k), 3.5 kbp (mEno1 3.5k) and 4k (mEno1 4k) reached, respectively, 1.9 times, 2.2 times and 2.4 times the value of that with the human EF1 α promoter.

[0189] It was found that the Chinese hamster-derived Eno1 gene promoter was also superior in the amount of antibody produced per cell (Qp) to the mouse-derived Eno1 gene promoter.

(Example 5) Study on Effect Brought about by
Combination of the Eno1 Gene Promoter Derived
from Each Species and A7 with Antibody
Expression Levels as Indicator in Fed-Batch
Culture

(5-1) Construction of Antibody Expression Vector

[0190] pDSLH3.1-Eno1-Y constructed in (2-1), and pDSLH3.1-Eno1-3.5-Y, pDSLH3.1-Eno1-2.8-Y, pDSLH3.1-mEno1-Y, pDSLH3.1-mEno1-3.5-Y, pDSLH3.1-mEno1-4.0-Y and pDSLH3.1-Eno1-4.0-Y constructed in (4-1) were used as parent vectors. The DNA element A7 (SEQ ID NO: 6) was inserted upstream of the expression cassettes of the vectors by a method similar to that in Patent Literature 5 to construct antibody expression vectors.

(5-2) Generation of Humanized Antibody Y-Expressing
Stable Pools

[0191] The CHO-01 cells were transfected with the parent vector and the antibody expression vector A7 according to the method described in (2-2). Drug selection culture was performed to generate a humanized antibody Y-expressing stable pool (two pools for each vector).

5-3) Evaluation of the Amount of Antibody Produced by
Fed-Batch Culture of Humanized Antibody Y-Expressing
Stable Pools

[0192] Fed-batch culture was performed in a 125 mL Erlenmeyer flask using each humanized antibody Y-expressing stable pool generated in (5-2). The basal medium used was G13, and the feed medium used was F13. The change in the number of viable cells and the change in the amount

of antibody produced are shown in FIGS. 5A and 5B, respectively. An average value of the amounts of antibodies produced in two pools generated for each vector, and the antibody productivities were compared. On day 14 of culture, the amount of antibody produced from the A7-containing antibody expression vector was 1.7 to 5.4 times the value of that from the A7-free antibody expression vector. Accordingly, it was found that either the Chinese hamster-derived or mouse-derived Eno1 gene promoter had greatly improved antibody productivity when used in combination with the DNA element A7.

Sequence Listing Free Text

[0193] SEQ ID NO: 1—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 3.0 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 2—Nucleotide sequence of the Mouse-derived Eno1 gene promoter from a nucleotide approximately 3.0 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 3—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 2.8 kbp upstream of the start codon of Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 4—Nucleotide sequence of Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 4.0 kbp upstream of the start codon of Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 5—Nucleotide sequence of DNA element A2 SEQ ID NO: 6—Nucleotide sequence of DNA element A7 SEQ ID NO: 7—Nucleotide sequence of DNA element A18 SEQ ID NO: 8—Primer for Eno1 gene promoter K24 SEQ ID NO: 9—Primer for Eno1 gene promoter K25 SEQ ID NO: 10—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 5.0

kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 11—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 4.5 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 12—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 3.5 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 13—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 2.7 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 14—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 2.6 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 15—Nucleotide sequence of the Chinese hamster-derived Eno1 gene promoter from a nucleotide approximately 2.5 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 16—Nucleotide sequence of the mouse-derived Eno1 gene promoter from a nucleotide approximately 4.0 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 17—Nucleotide sequence of the mouse-derived Eno1 gene promoter from a nucleotide approximately 3.5 kbp upstream of the start codon of the Eno1 gene to the nucleotide immediately upstream of the nucleotide sequence corresponding to the start codon SEQ ID NO: 18—Differential sequence between SEQ ID NO: 3 (Eno1 2.8 k) and SEQ ID NO: 13 (Eno1 2.7 k)

SEQUENCE LISTING

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SEQ ID NO: 12 moltype = DNA length = 3500
 FEATURE Location/Qualifiers
 source 1..3500
 mol_type = genomic DNA
 organism = Cricetulus griseus

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SEQ ID NO: 13 moltype = DNA length = 2707
 FEATURE Location/Qualifiers
 source 1..2707
 mol_type = genomic DNA
 organism = Cricetulus griseus

SEQUENCE: 13

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SEQ ID NO: 14 moltype = DNA length = 2607
 FEATURE Location/Qualifiers
 source 1..2607
 mol_type = genomic DNA
 organism = Cricetulus griseus

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FEATURE            Location/Qualifiers
source              1..2507
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                    organism = Cricetulus griseus

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SEQ ID NO: 16 moltype = DNA length = 4000
 FEATURE Location/Qualifiers
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 organism = Mus musculus

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1. A polynucleotide that is a Chinese hamster-derived gene promoter, the polynucleotide comprising a nucleotide sequence that is selected from SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13 and SEQ ID NO: 14, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.

2. The polynucleotide according to claim 1, which comprises a nucleotide sequence that is selected from SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11 and SEQ ID NO: 12, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.

3.-10. (canceled)

11. A polynucleotide that is a mouse-derived gene promoter, the polynucleotide comprising a nucleotide sequence that is selected from SEQ ID NO: 2, SEQ ID NO: 17 and SEQ ID NO: 16, or has 85% or higher sequence identity to each of the sequences, or a partial sequence of the nucleotide sequence.

12. The polynucleotide according to claim 11, which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 2.

13. The polynucleotide according to claim 11, which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 17.

14. The polynucleotide according to claim 11, which consists of a nucleotide sequence having 99% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 16.

15. (canceled)

16. A polynucleotide comprising a nucleotide sequence having 95% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 2, SEQ ID NO: 17 or SEQ ID NO: 16, or a partial sequence of the nucleotide sequence, the polynucleotide having promoter activity.

17. A polynucleotide that hybridizes under stringent conditions to a polynucleotide consisting of a nucleotide sequence complementary to the nucleotide sequence of SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 2, SEQ ID NO: 17 or SEQ ID NO: 16, the polynucleotide having promoter activity.

18. The polynucleotide according to claim 17, which is a Chinese hamster-derived gene promoter and comprises a nucleotide sequence of SEQ ID NO: 18, or a nucleotide sequence having 85% or higher sequence identity to the sequence.

19. A foreign gene expression unit comprising the polynucleotide according to claim 1 and a foreign gene.

20. The foreign gene expression unit according to claim 19, wherein the foreign gene is a gene encoding a protein.

21. The foreign gene expression unit according to claim 20, wherein the foreign gene is a gene encoding a heteromultimeric protein.

22. The foreign gene expression unit according to claim 21, wherein the foreign gene is a gene encoding an antibody or an antigen-binding fragment thereof.

23. A foreign gene expression vector comprising the foreign gene expression unit according to claim 19.

24. A foreign gene expression vector comprising the foreign gene expression unit according to claim 19, and one or more polynucleotides selected from the following (a) to (e):

- (a) a polynucleotide comprising the nucleotide sequence of SEQ ID NO: 5, or a partial sequence thereof;
- (b) a polynucleotide comprising the nucleotide sequence of SEQ ID NO: 6, or a partial sequence thereof;
- (c) a polynucleotide comprising the nucleotide sequence of SEQ ID NO: 7, or a partial sequence thereof;
- (d) a polynucleotide comprising a nucleotide sequence having 80% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 5, SEQ ID NO: 6 or SEQ ID NO: 7, or a partial sequence of the nucleotide sequence, the polynucleotide having foreign gene expression-enhancing activity; and
- (e) a polynucleotide comprising a nucleotide sequence having 85% or higher sequence identity to the nucleotide sequence of SEQ ID NO: 5, SEQ ID NO: 6 or SEQ ID NO: 7, or a partial sequence of the nucleotide sequence, the polynucleotide having foreign gene expression-enhancing activity.

25. A transfected cell into which the foreign gene expression vector according to claim 23 has been introduced.

26. The transfected cell according to claim 25, wherein the cell is a cultured cell derived from a mammal.

27. The transfected cell according to claim 26, wherein the cell is a COS-1 cell, a 293 cell, or a CHO cell.

28. A method for producing a foreign gene-derived protein, comprising culturing the transfected cell according to claim 25, and obtaining the foreign gene-derived protein from the culture.

29. The production method according to claim 28, wherein the foreign gene-derived protein is a monomeric protein or a multimeric protein.

30. The production method according to claim **29**, wherein the multimeric protein is a heteromultimeric protein.

31. The production method according to claim **30**, wherein the heteromultimeric protein is an antibody protein.

32-33. (canceled)

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